



WELCOME

SYSTEM ADMIN AND SECURITY

Course: COMP 175-01 (CRN 80249)

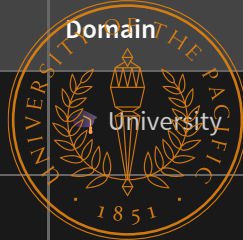
Term: Fall 2026 (Aug 24 – Dec 11, 2026)

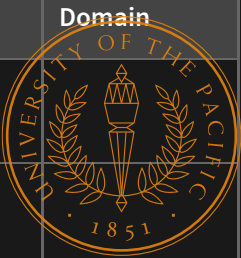
Time: 8:25 AM – 9:45 AM on Tuesday, Thursday

Location: CTC 214

Credits: 3.000

Office Hours: Tuesday, Thursday 1:30–2:30 PM, CTC 117 or [Zoom](#)

Week	Dates	Topic	Notes	Assignment	Domain
Week 1	Aug 25, Aug 27	Virtualization & Isolation	Commands: wsl , sudo , apt , pacman , nmap , whois , dig , VBoxManage	Lab/HW 1	 University
Week 2	Sep 1, Sep 3	Command-Line & Permissions	Sep 7 is Labor Day.		
Week 3	Sep 8, Sep 10	File Systems & Storage		Lab/HW 2	
Week 4	Sep 15, Sep 17	Access Control & Auditing			
Week 5	Sep 22, Sep 24	Process Mgmt & Remote Access		Lab/HW 3	
Week 6	Sep 29, Oct 1	Service Management		Project Part 1 (Individual)	 Automotive
Week 7	Oct 6, Oct 8	Network Configuration		Lab/HW 4	
Week 8	Oct 13, Oct 15	Name Resolution & Secure Transport		Midterm Exam: Oct 15 (in class)	
Week 9	Oct 20, Oct 22	Logging & Security Monitoring	Project Part 1 Presentation	Lab/HW 5, Project Part 2 (Team)	
Week 10	Oct 27, Oct 29	Time Synchronization			
Week 11	Nov 3, Nov 5	Automation Scripting		Lab/HW 6	 Hospital
Week 12	Nov 10, Nov 12	Firewalls & Network Security	Project Part 2 Presentation		
Week 13	Nov 17, Nov 19	Deployment & Network Segmentation			

Week	Dates	Topic	Notes	Assignment	Domain
Week 14	Nov 24, Nov 26	Containerization	No class Nov 26 (Thanksgiving)		
Week 15	Dec 1, Dec 3	Infrastructure as Code	Project Showcase: Fri, Dec 4, 2–3:15 PM, Makerspace		
Week 16	Dec 8, Dec 10	Final Presentations & Course Wrap-up		Final Exam: Dec. 10th, 8–10AM in CTC 214	

Email: sberhe@pacific.edu

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Office Hours: Tue/Thu 1:30–2:30 PM, CTC 117

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[Zoom link](#)

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[Syllabus: Canvas](#)



MODULE 1

WELCOME & INTRODUCTION: VIRTUALIZATION



AGENDA

1. The sysadmin role
2. Linux vs. other OSes
3. Shells: cmd, PowerShell & Bash
4. Hands-on: install Linux
5. University Domain: Plan VM
6. What is virtualization
7. VirtualBox & WSL
8. Real-world risks
9. Lab 1 & Homework 1

A screenshot of a PowerShell terminal window. The title bar says "PowerShell". The prompt is "PS C:\Users\John>". The user has entered "\$PSVersionTable", which has outputted a table of system information. The user has then entered a function definition for "Say-Hello" and called it. The output of the function is "Hello, World!".

```
PowerShell 7.5.0
PS C:\Users\John> $PSVersionTable

Name                           Value
----                           -
PSVersion                      7.5.0
PSEdition                      Core
GitCommitId                    7.5.0
OS                              Microsoft Windows 10.0.22631
Platform                       Win32NT
PSCompatibleVersions           {1.0, 2.0, 3.0, 4.0...}
BuildNumber                     22H2
CLRVersion                     4.0.30319.494
SerializationVersion           1.1.0.1
WSManStackVersion              3.0

PS C:\Users\John> function Say-Hello {
>> Write-Host "Hello, World!"
>> }
PS C:\Users\John> Say-Hello
Hello, World!
PS C:\Users\John> |
```

PowerShell (Source: [Wikimedia Commons](#))



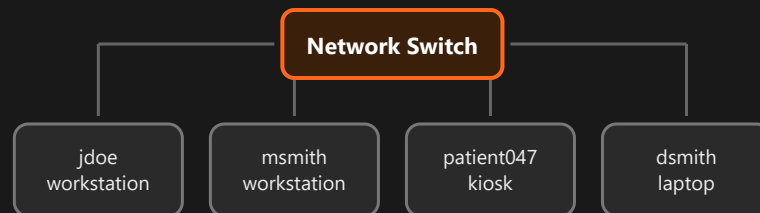
THE SYSADMIN ROLE





WHAT DOES A SYSTEM **ADMIN** & SECURITY DO?

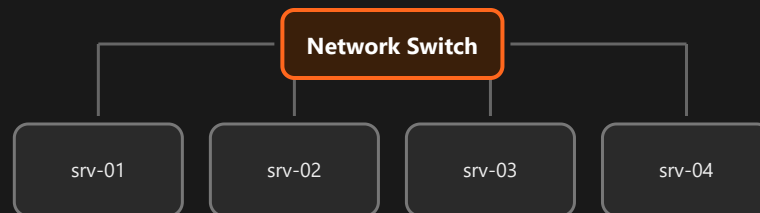
- Manages the users network
- Workstations, kiosks & laptops
- Connected via network switch





WHAT DOES A SYSTEM **ADMIN** & SECURITY DO?

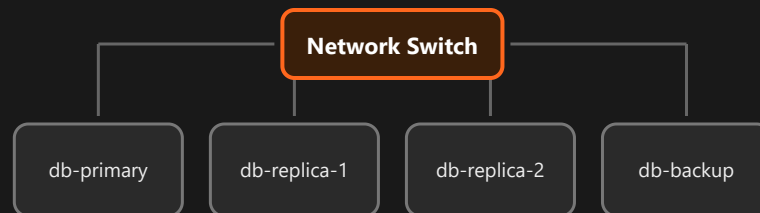
- Manages the machines network
- Servers behind a switch
- Foundation for services





WHAT DOES A SYSTEM **ADMIN** & SECURITY DO?

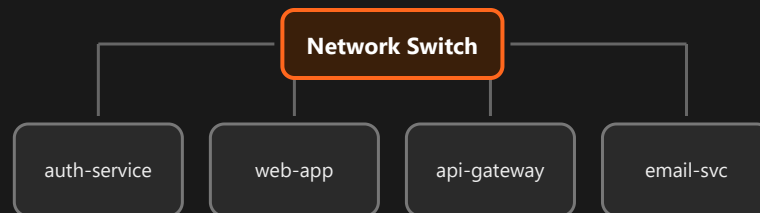
- Manages the databases network
- Primary, replicas & backup
- Keeps data available





WHAT DOES A SYSTEM **ADMIN** & SECURITY DO?

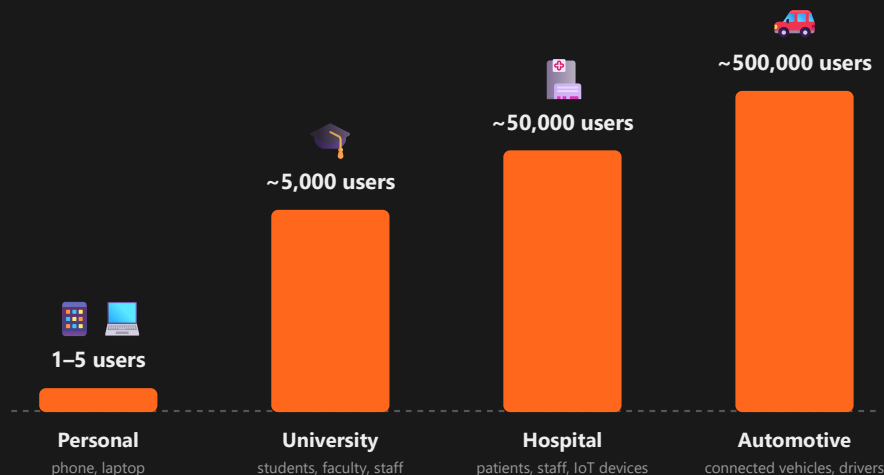
- Manages the software network
- Auth, web & API services
- Keeps applications running





WHAT DOES A SYSTEM **ADMIN** & SECURITY DO?

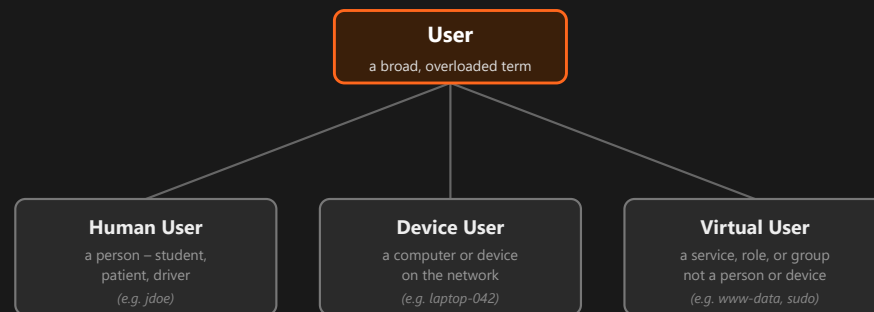
- Scales from 1 to 500,000 users
- Personal, campus & hospital
- Same skills, bigger scale





WHAT DOES A SYSTEM **ADMIN** & SECURITY DO?

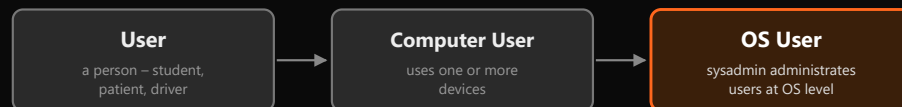
- Manages three kinds of user
- Human, device & virtual
- Each needs its own account





WHAT DOES A SYSTEM **ADMIN** & SECURITY DO?

- A person becomes a user
- Then a computer user
- Then an OS user account



OS USER ACCOUNTS

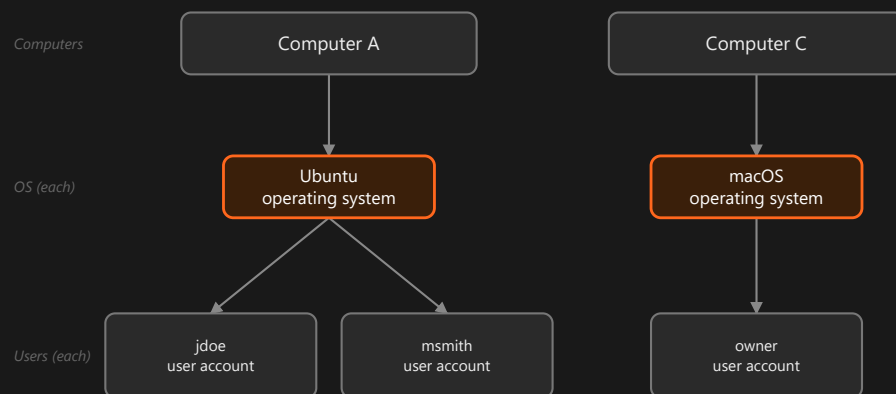
```
$ cat /etc/passwd | grep -E 'alex|jdoe|icu|veh'
alex:x:1000:1000:Personal,,,:/home/alex:/bin/bash
jdoe:x:1001:1001:Student,,,:/home/jdoe:/bin/bash
icu047:x:1002:1002:Hospital,,,:/home/icu047:/bin/rbash
veh3210:x:1003:1003:Vehicle,,,:/home/veh3210:/bin/sh
```

Copy



WHAT DOES A SYSTEM **ADMIN** & SECURITY DO?

- Two computers, one OS each
- Ubuntu & macOS shown here
- Each OS has its own users





WHAT DOES A SYSTEM ADMIN & **SECURITY** DO?

- Builds & defends systems
- Tests for weaknesses
- Works across all three





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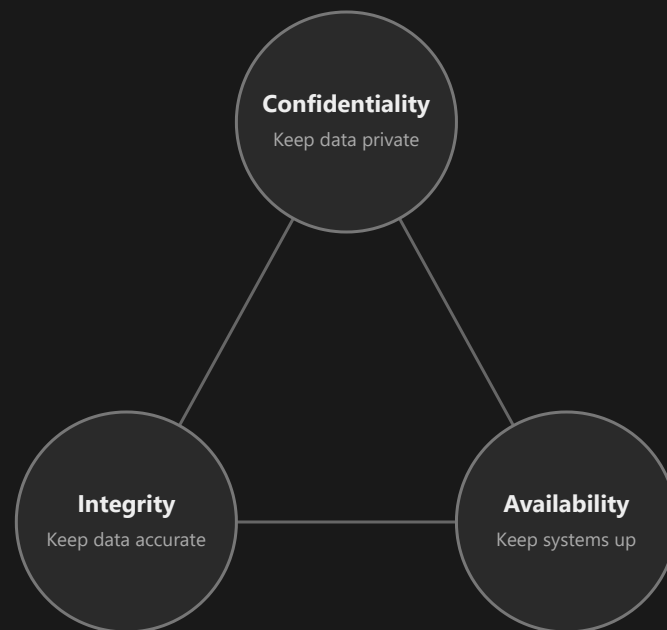


⚠ Without Purple Team collaboration, Red and Blue often **repeat** each other's mistakes.



WHAT DOES A SYSTEM **ADMIN** & **SECURITY** DO?

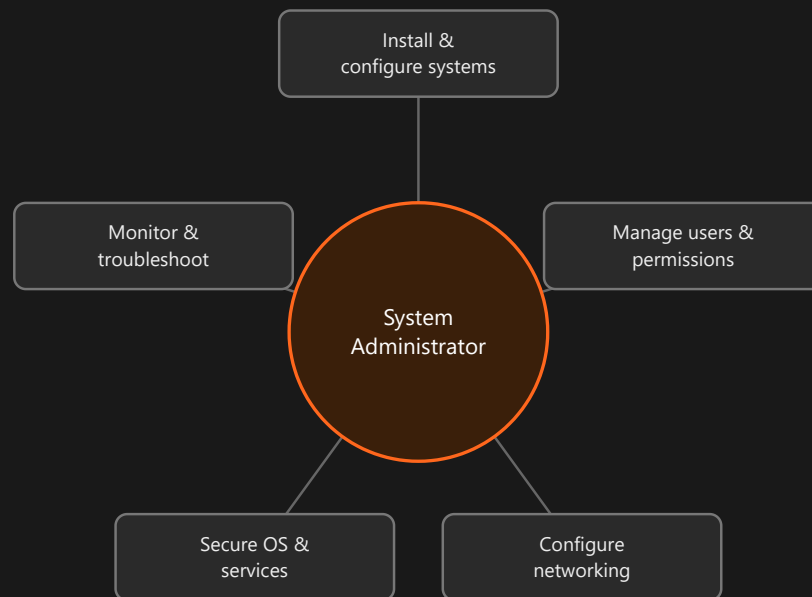
- Confidentiality: private data
- Integrity: accurate data
- Availability: systems stay up





WHAT DOES A **SYSTEM** ADMIN & SECURITY DO?

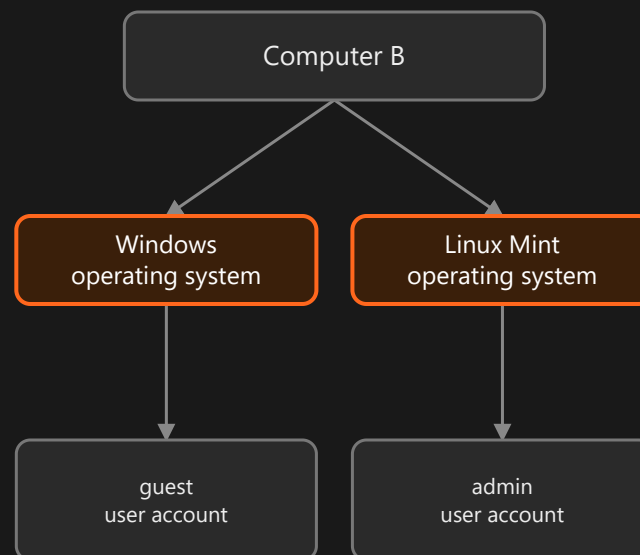
- Installs & configures systems
- Manages users & permissions
- Monitors & troubleshoots





WHAT DOES A **SYSTEM** ADMIN & SECURITY DO?

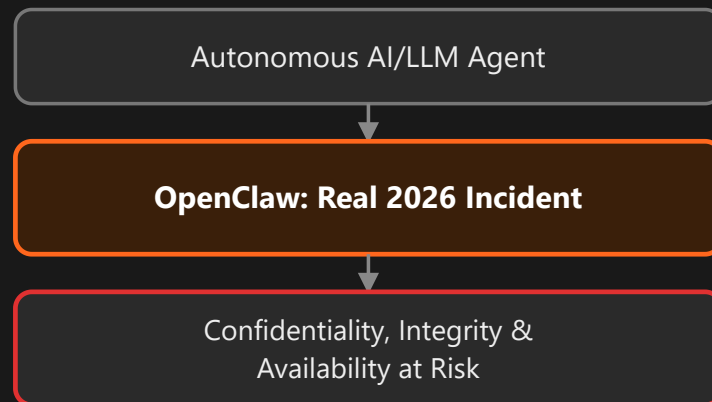
- One computer, two systems
- Windows & Linux Mint installed
- Each system has its own users





WHAT DOES A SYSTEM ADMIN & SECURITY NOT DO? (AI/LLM)

- Never runs AI/LLM unsupervised
- Risks confidentiality
- Risks integrity & availability
- OpenClaw (2026): real incident

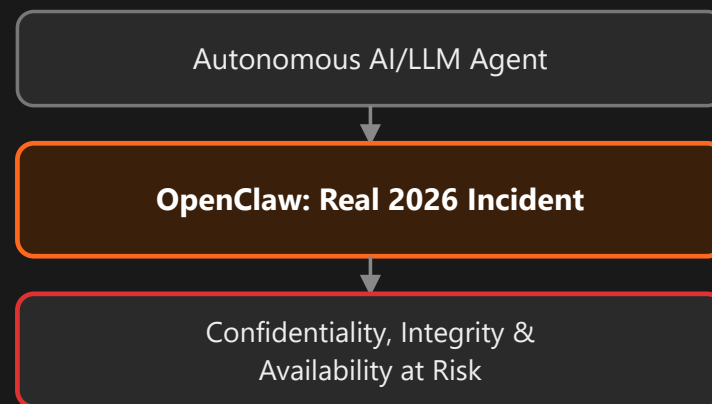


“Researchers Expose OpenClaw AI Agent Credential Leak During Phishing Simulation” (2026) (Source: [Cyber Press](#))



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- Risks integrity & availability
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“Researchers Expose OpenClaw AI Agent Credential Leak During Phishing Simulation” (2026) (Source: [Cyber Press](#))

🚨 OpenClaw is a real autonomous AI agent given shell access and credentials. In a 2026 phishing test, one company's OpenClaw agent handed over a customer database and forwarded AWS keys in **plaintext**.



WHAT DOES A SYSTEM ADMIN & SECURITY NOT DO? (USER CONTENT)

- Doesn't own user data
- Doesn't manage user software
- Admin manages the platform

Admin Manages

Operating System
Network & Access
User Accounts

User Owns

Their Files
Their Data
Their Software

“Court Rules that Stored Communications Act Applies to Former Employee Emails” (2013) (Source: [CloudNine](#))



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“Court Rules that Stored Communications Act Applies to Former Employee Emails” (2013) (Source: [CloudNine](#))

⚠ In *Lazette v. Kulmatycki* (2013), a supervisor read **48,000** of a former employee's personal emails over 18 months after she returned her company phone.



WHAT DOES A SYSTEM ADMIN & SECURITY NOT DO? (APP CODE)

- Doesn't write app code
- That's the developer's job
- Admin runs & secures it



“The \$440 Million Software Error at Knight Capital” (2012)
(Source: [Henrico Dolfing](#))



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“The \$440 Million Software Error at Knight Capital” (2012)
(Source: [Henrico Dolfing](#))

🚨 In 2012, Knight Capital lost **\$460 million** in 45 minutes after a technician's manual deployment left old code running on one server.



WHAT DOES A SYSTEM ADMIN & SECURITY NOT DO? (SNOOPING)

- Root access isn't a license
- Doesn't snoop on private data
- Ethics, not just capability



Robbins v. Lower Merion School District (2010) (Source: [Wikipedia](#))



WHAT DOES A SYSTEM ADMIN & SECURITY NOT DO? (SNOOPING)

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Robbins v. Lower Merion School District (2010) (Source: [Wikipedia](#))

🚨 In the Lower Merion webcam case (2010), a network technician secretly activated laptop cameras, leading to a **\$610,000** settlement.



WHAT DOES A SYSTEM ADMIN & SECURITY DO? (RACI)

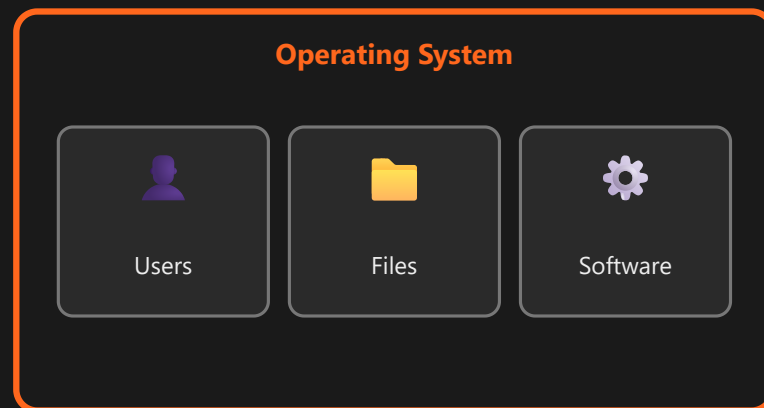
- R: Responsible, does the work
- A: **Accountable**, owns outcome
- C: Consulted, gives input
- I: Informed, kept in loop

Task	Admin	Dev	Sec	User
Patching	R/ A	I	C	I
App Code	I	R/ A	C	-
Access Review	R	I	A	C
Incident Resp.	R	I	A	I
AI/LLM Use	R	C	A	I
User Data	I	C	C	R/ A
Network Setup	R/ A	I	C	I
Backups & Snapshots	R/ A	I	C	I
System Uptime	R/ A	I	C	I
VM Configuration	R/ A	I	I	I



IF A SYSADMIN DOESN'T WRITE APP CODE AND DOESN'T OWN USER DATA, WHAT'S LEFT TO MANAGE?

- Manages user accounts
- Manages files & storage
- Manages installed software

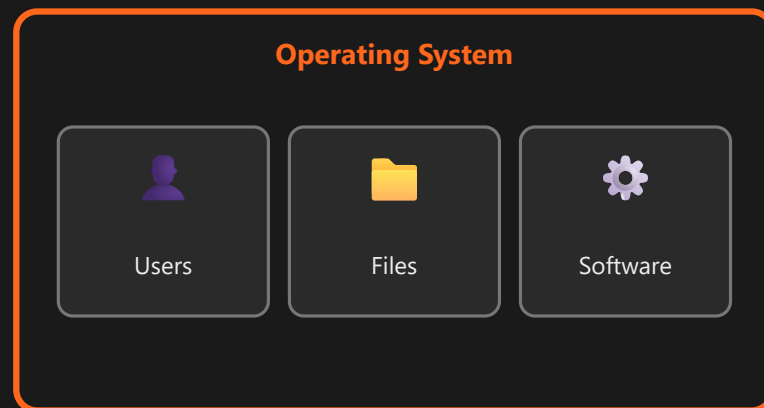




IF A SYSADMIN DOESN'T WRITE APP CODE AND DOESN'T OWN USER DATA, WHAT'S LEFT TO MANAGE?

- Manages user accounts
- Manages files & storage
- Manages installed software

→ **The Operating System**.





QUESTIONS?



LINUX VS. OTHER OSES



HOW MANY OPERATING SYSTEMS EXIST?

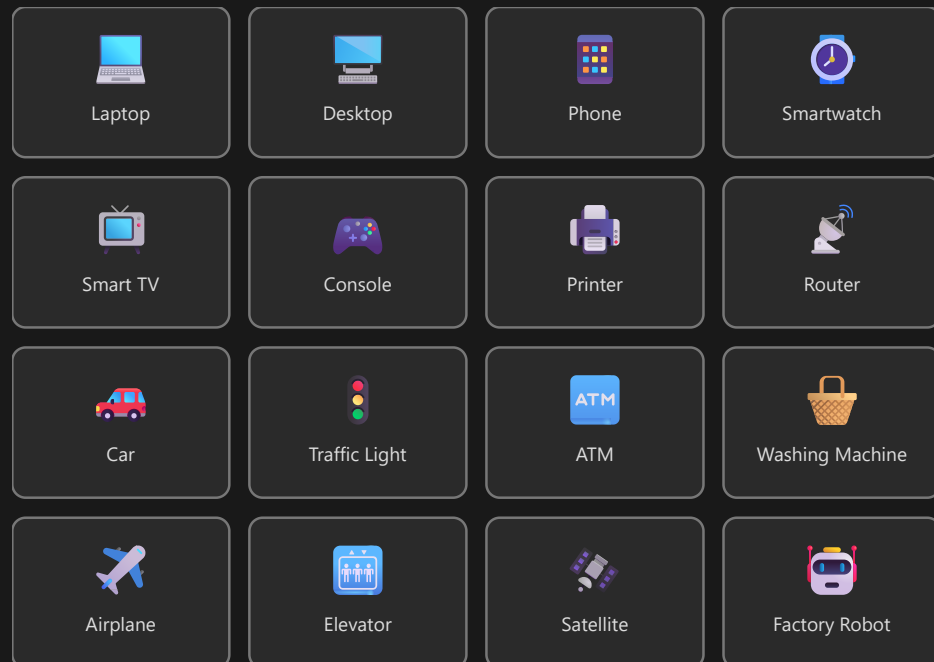
- 50+ real names shown here
- Desktop, mobile & embedded
- Many trace to Unix or Linux

Windows 11 **macOS Sequoia** **Ubuntu** **Android**
iOS **Debian** **Fedora** **Windows Server 2025**
RHEL Linux Mint CentOS Stream Arch Linux openSUSE
FreeBSD Chrome OS Windows 10 **Kali Linux**
Alpine Linux Manjaro Pop!_OS Rocky Linux AlmaLinux
Solaris AIX HP-UX iPadOS watchOS Raspberry Pi OS
Windows Server 2022 SUSE Linux Enterprise Tails Qubes OS
FreeDOS ReactOS Haiku OpenBSD NetBSD Slackware Gentoo Void Linux
NixOS Zorin OS elementary OS MX Linux Deepin Puppy Linux
Tiny Core Linux tvOS VxWorks QNX



WHERE IS AN OPERATING SYSTEM INSTALLED?

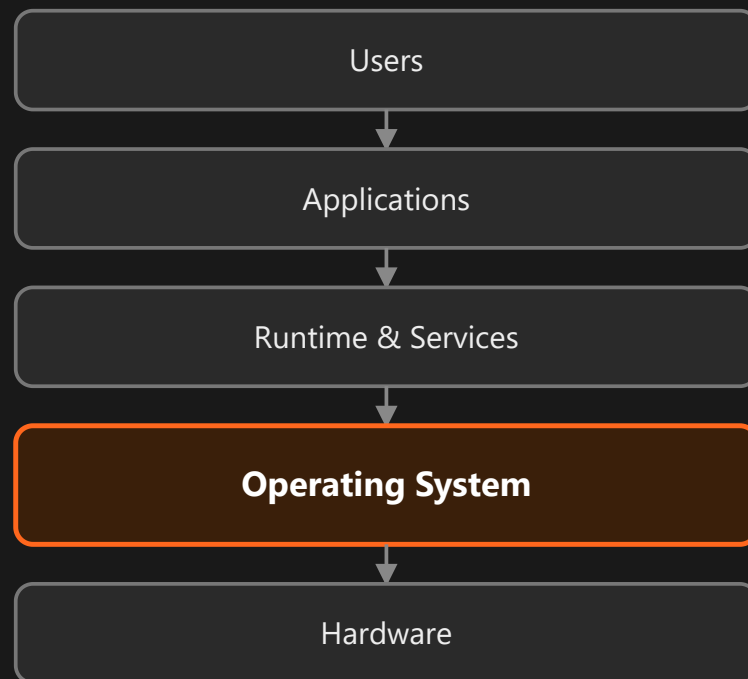
- Not just PCs & phones
- Cars, ATMs, appliances too
- e.g., embedded Linux in cars





WHAT SITS AT THE BASE OF EVERY SYSTEM?

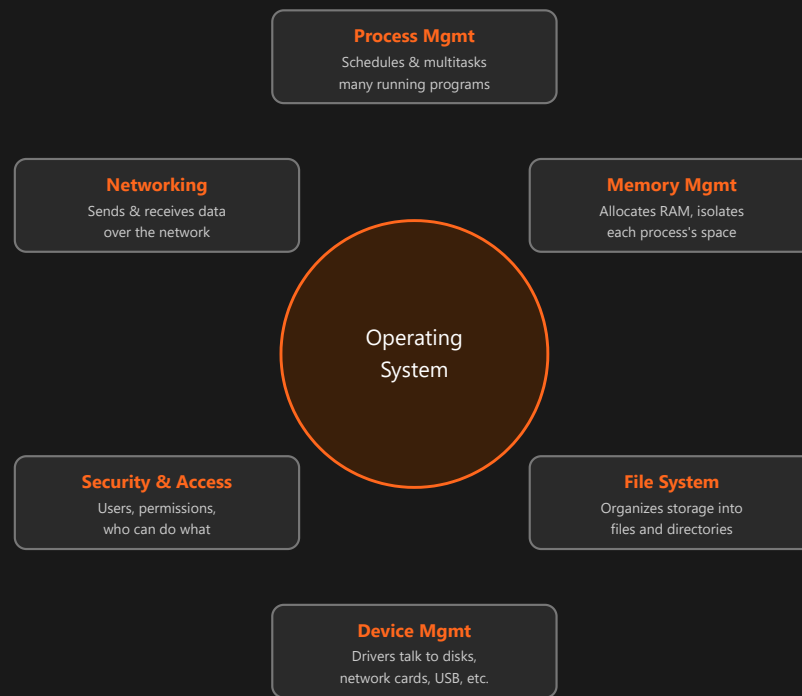
- Every layer depends on it
- Apps can't bypass it
- e.g., Windows 11 on laptops





WHAT DOES AN OS ACTUALLY DO?

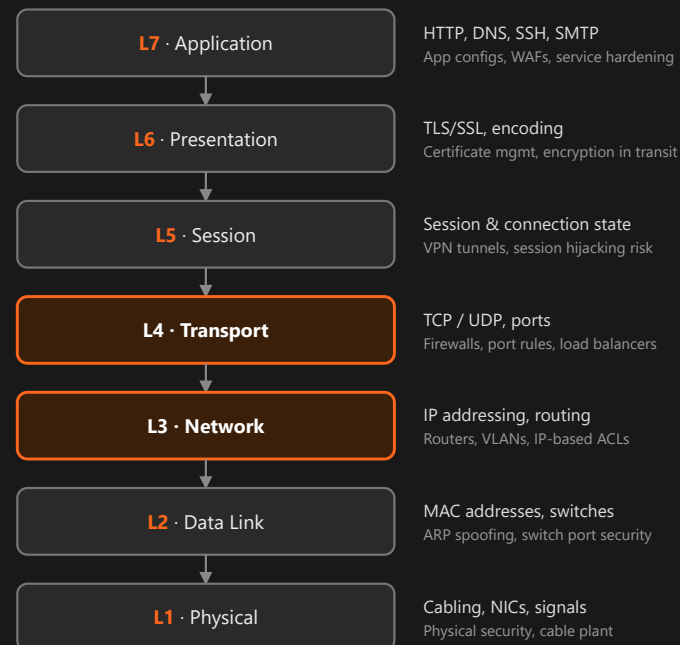
- Manages memory & processes
- Manages files & networking
- Enforces security & access





WHERE DOES ADMIN & SECURITY WORK HAPPEN?

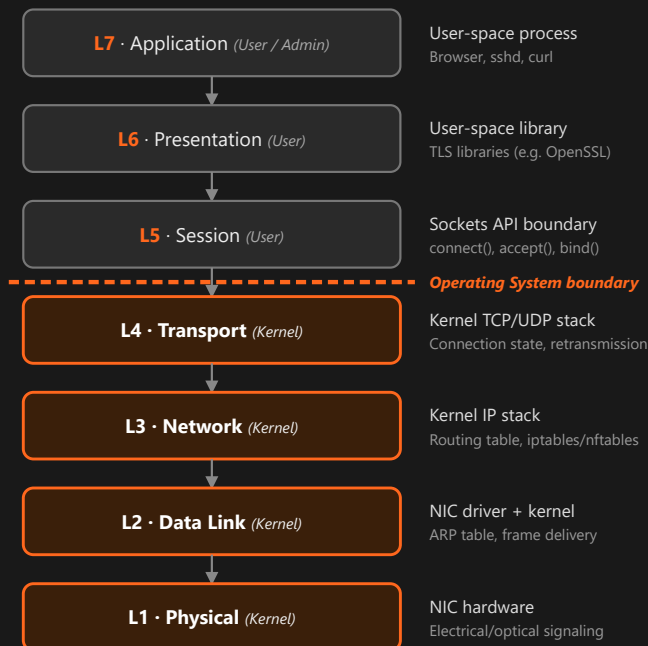
- Across all 7 OSI layers
- From cabling to app configs
- e.g., Firewall rules (L4)





WHERE DOES THE OS FIT IN THE NETWORK STACK?

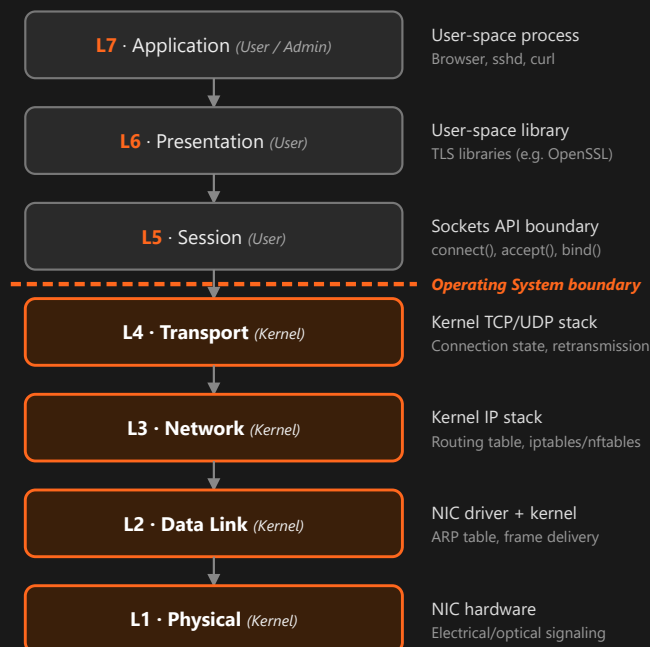
- Sits at Layer 7, technically
- root has elevated privilege
- e.g., sshd on a campus server





WHERE DOES THE OS FIT IN THE NETWORK STACK?

- Sits at Layer 7, technically
- root has elevated privilege
- e.g., sshd on a campus server



Admin isn't its own layer – admin tools (iptables, systemctl, ip) run at L7 like any app, but with elevated (**root**) privilege to reach across the boundary and reconfigure L4–L1 kernel state directly, instead of only using the unprivileged sockets API like an ordinary app.



UNIX

Official UNIX trademark logo (Source: [Wikipedia](#), [Unix](#))

- Born 1969, Bell Labs (AT&T)
- Thompson & Ritchie
- Direct descendants: Solaris,
- AIX, HP-UX, macOS (Darwin)
- "UNIX" is a trademark, not
- a free label – certified only



LINUX

Tux, the Linux mascot (Source: [Wikipedia](#), [Linux](#) (illustration by Larry Ewing))

- Born 1991, Linus Torvalds
- A new kernel, Unix-like
- Not built from Unix source
- Free & open source (GPL)
- Paired with GNU tools to
- form a full OS



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Don't call Linux "Unix" – wrong, and a trademark violation. Say "Unix-like" instead.



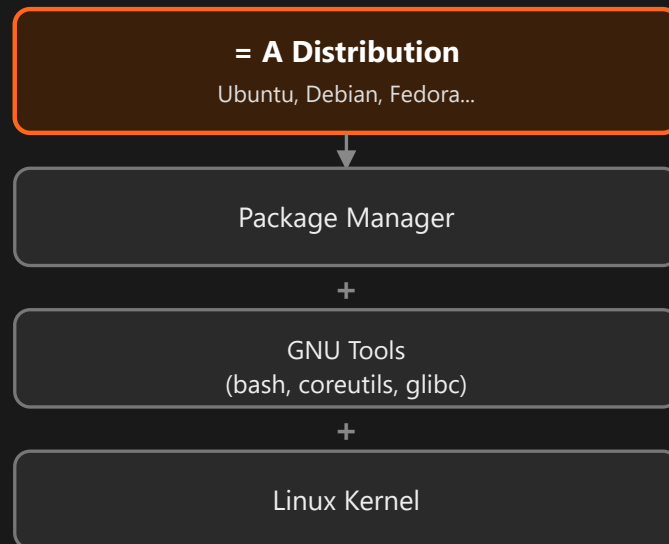
WHAT'S THE DIFFERENCE: KERNEL VS. DISTRO?

LINUX (THE KERNEL)

- Just the kernel Torvalds wrote
- Manages CPU, memory, devices
- By itself, not usable

A DISTRIBUTION

- Kernel + GNU tools + pkg mgr
- + sane defaults, installer
- Ubuntu, Debian, Fedora, etc.





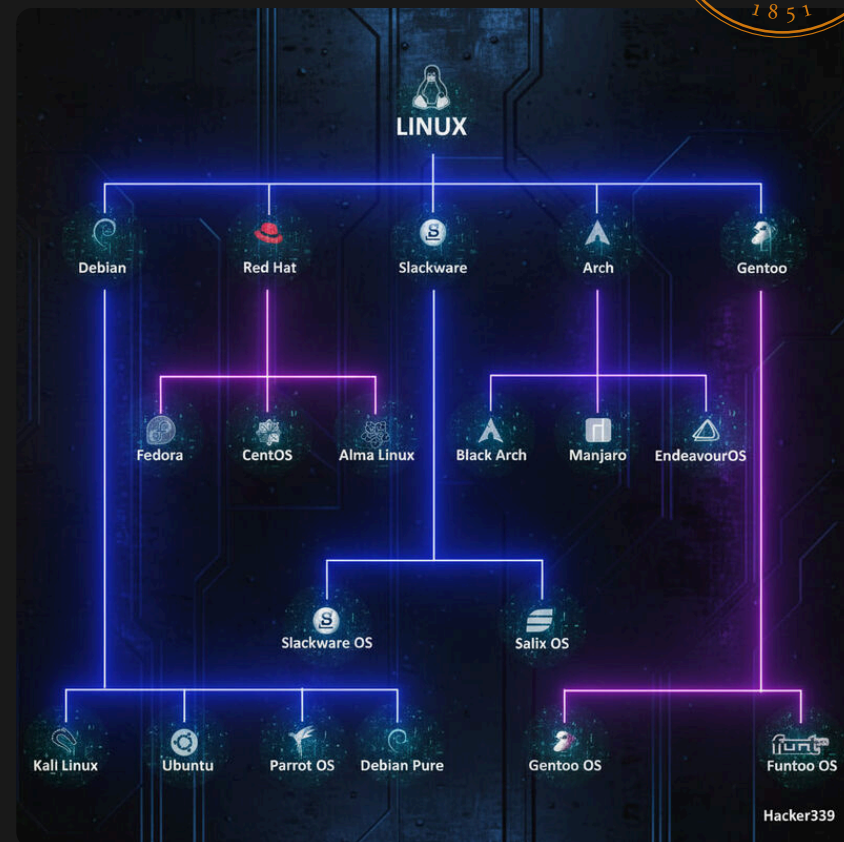
FROM OS CONCEPTS TO LINUX

- Concepts apply to any OS
- Now: Linux, specifically
- e.g., Unix-like kernel



HOW DID LINUX DIVERGE INTO MANY DISTROS?

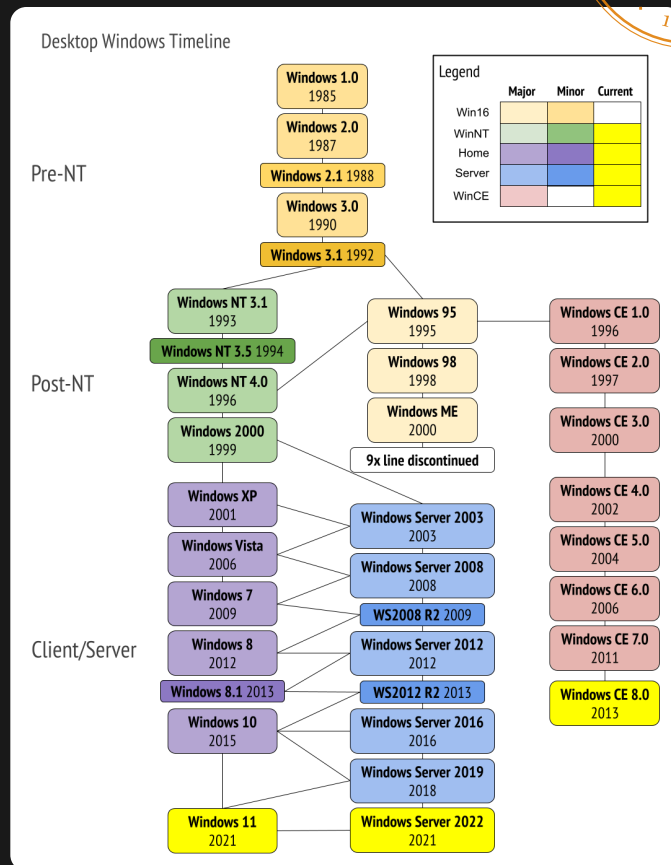
- One kernel, many descendants
- Debian, Red Hat, Arch branch
- e.g., Kali forked from Debian



Linux distro family tree (Source: [LinkedIn](#), [Hacker339](#))

HOW DID WINDOWS STAY ONE LINE?

- One vendor, one line
- NT branch replaced 9x line
- e.g., Windows 11 from NT

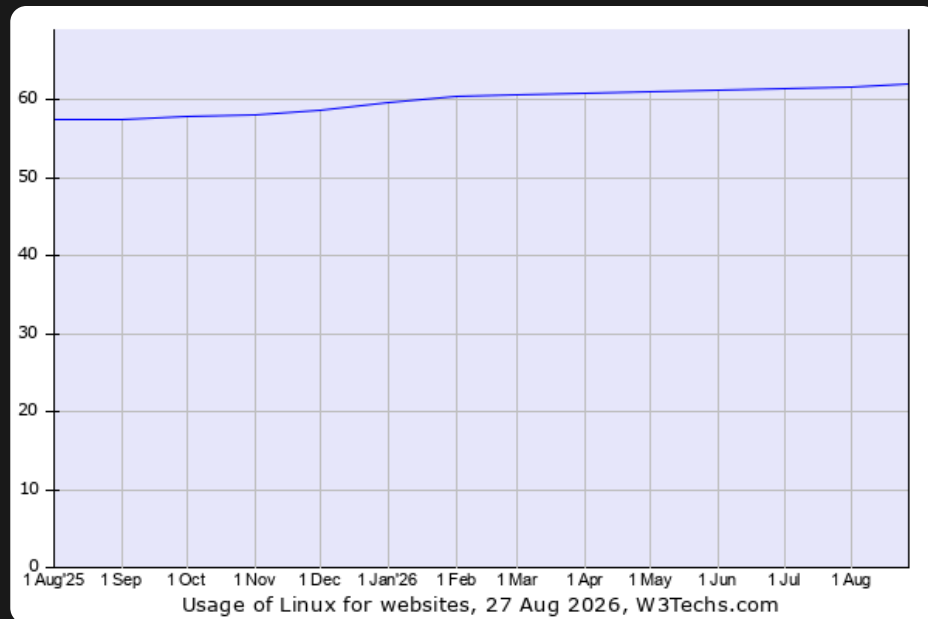


Windows version history (Source: [Wikimedia Commons](https://commons.wikimedia.org/wiki/File:Windows_versions_timeline.png))



LINUX'S GROWTH OVER TIME

- 57.5% of websites, Aug 2025
- 62.1% of websites, Aug 2026
- Steady growth, no plateau



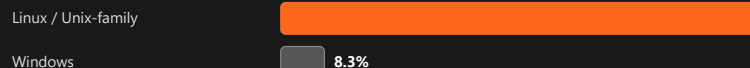
Linux usage share for websites, 2025–2026 (Source: [W3Techs](https://w3techs.com), live server-side data, Aug 2026)



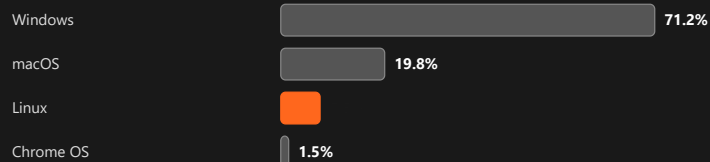
WHICH OS RUNS WHERE?

- Servers lean Linux (91.9%)
- Desktops lean Windows (71.2%)
- Phones lean Android/Linux

Servers (web-facing, W3Techs, Aug 2026)



Desktop & Laptop (StatCounter, Jul 2026)



Phones (StatCounter, Jul 2026)

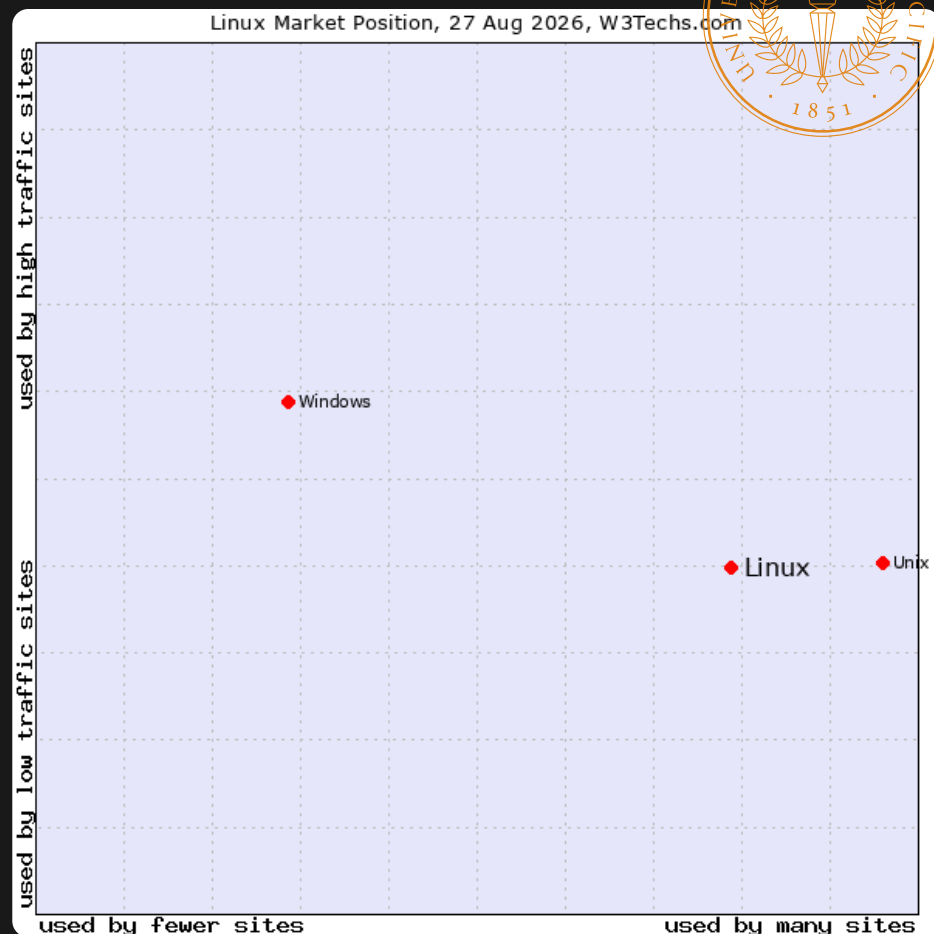


Sources: [W3Techs](#), [StatCounter](#)



LINUX'S MARKET POSITION

- Used by low- and high-traffic sites about equally
- Windows skews toward lower-traffic sites

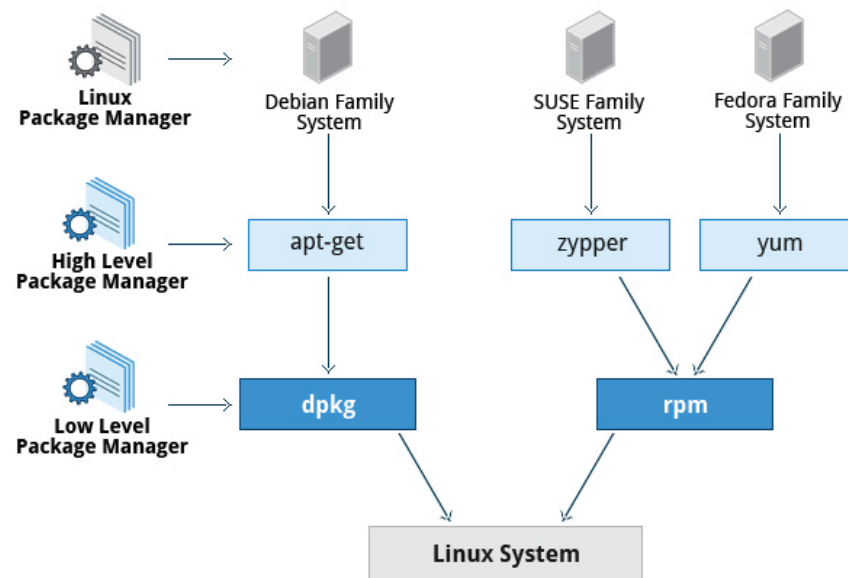


Linux market position by site traffic (Source: [W3Techs](https://w3techs.com), live server-side data, Aug 2026)



WHERE DO YOU FIND & INSTALL PACKAGES (SOFTWARE)?

- Official repos (~70K pkgs)
- Flatpak (~3K), Snap (~6K)
- e.g., apt install curl (WSL)
- e.g., apt-get remove curl



Package managers by Linux distribution (Source: [Devopedia](#), [Package Manager](#))



HOW DO YOU INSTALL WITH APT?

- Debian, Ubuntu format (.deb)
- e.g., apt install mediawiki
- Won't work with pacman

```
wikipedia@wikimedia.org:~$ sudo apt-get install mediawiki
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libkpi7 libkxiv2-8 python-urwid python-iniparse python-wicd libtelepathy-qt4-0
  libkdcraw8
Use 'apt-get autoremove' to remove them.
The following extra packages will be installed:
  libdbd-mysql-perl libdbi-perl libhtml-template-perl libnet-daemon-perl liblprc-perl
  mysql-client-5.1 mysql-server mysql-server-5.1 php5
Suggested packages:
  libipc-sharedcache-perl mediawiki-math memcached clamav tinycs mailx
The following NEW packages will be installed:
  libdbd-mysql-perl libdbi-perl libhtml-template-perl libnet-daemon-perl liblprc-perl
  mediawiki mysql-client-5.1 mysql-server mysql-server-5.1 php5
0 upgraded, 10 newly installed, 0 to remove and 223 not upgraded.
Need to get 27.8MB of archives.
After this operation, 84.6MB of additional disk space will be used.
Do you want to continue [Y/n]? █
```

apt: Debian, Ubuntu (Source: [Wikimedia Commons](#))



HOW DO YOU INSTALL WITH PACMAN?

- Arch Linux's package manager
- e.g., `pacman -S mediawiki`
- Won't work with `apt`

```
root@upho-lsde-miral:~# sudo pacman -Syu
:: Synchronizing package databases...
core is up to date
extra is up to date
community is up to date
multilib is up to date
home_linux/linux_chromeos_Arch is up to date
:: Starting full system upgrade...
Resolving dependencies...
Looking for conflicting packages...

Packages (6) model4-16.0-1 plasma-wayland-session-5.22.5-2 plasma-workspace-5.22.5-2 python-reportlab-3.6.1-1 qt5-base-5.15.2-kde-r222-2
qt5-wayland-5.15.2-kde-r201

Total Download Size: 35.68 MiB
Total Installed Size: 158.48 MiB
Net Upgrade Size: 0.15 MiB

:: Proceed with installation? [Y/n] Y
:: Retrieving packages...
qt5-wayland-5.15.2-kde-r201_04 1123.9 KiB 183 KiB/s 00:00
plasma-wayland-session-5.22.5-2_04_04 0.3 KiB 12.3 KiB/s 00:00
python-reportlab-3.6.1-1_04_04 2.5 MiB 248 KiB/s 00:11
model4-16.0-1_04_04 8.8 MiB 256 KiB/s 00:35
plasma-workspace-5.22.5-2_04_04 10.4 MiB 219 KiB/s 00:39
qt5-base-5.15.2-kde-r222-2_04_04 11.8 MiB 321 KiB/s 00:42
qt5-wayland-5.15.2-kde-r201_04 15.8 MiB 877 KiB/s 00:42

(4/6) checking keys in keyring
(4/6) checking package integrity
(4/6) loading package files
(4/6) checking for file conflicts
(4/6) checking available disk space
:: Running pre-transaction hooks...
(1/2) Checking arch news with Intranet ...
There are no unread news items
(1/2) Retrieving newer snapshots for the following configurations ...
-- root:
:: Processing package changes...
(1/6) upgrading model4
(2/6) upgrading qt5-base
(3/6) upgrading qt5-wayland
(4/6) upgrading plasma-workspace
(5/6) upgrading plasma-wayland-session
(6/6) upgrading python-reportlab
:: Running post-transaction hooks...
(1/6) Purging transactional-metadata...
(2/6) Refreshing PackageKit...
(3/6) Reloading Systemd Unit configuration...
(4/6) Updating icon theme caches...
(5/6) Updating the desktop file MIME type cache...
(6/6) Performing wrapper post snapshots for the following configurations...
-- root: 0
```

pacman: Arch Linux (Source: [Wikimedia Commons](#))



WHERE DOES "PACKAGE" COME FROM?

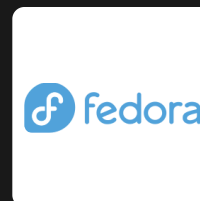
- 1530s: pack + -age (bundle)
- 1990s: Linux borrows term
- Same shipping metaphor





MAIN DISTRIBUTIONS

Distro	Pkg Mgr	Release	Best For
Debian	apt (.deb)	Slow, stable	Servers
Ubuntu	apt (.deb)	Fixed + LTS	Beginners, cloud
Fedora	dnf (.rpm)	~6 months	Newest features
Arch	pacman	Rolling	Full control, DIY



Debian, Ubuntu, Fedora & Arch logos (Source: official project assets (debian.org, Wikipedia, archlinux.org))



WHERE DOES "DISTRO" COME FROM?

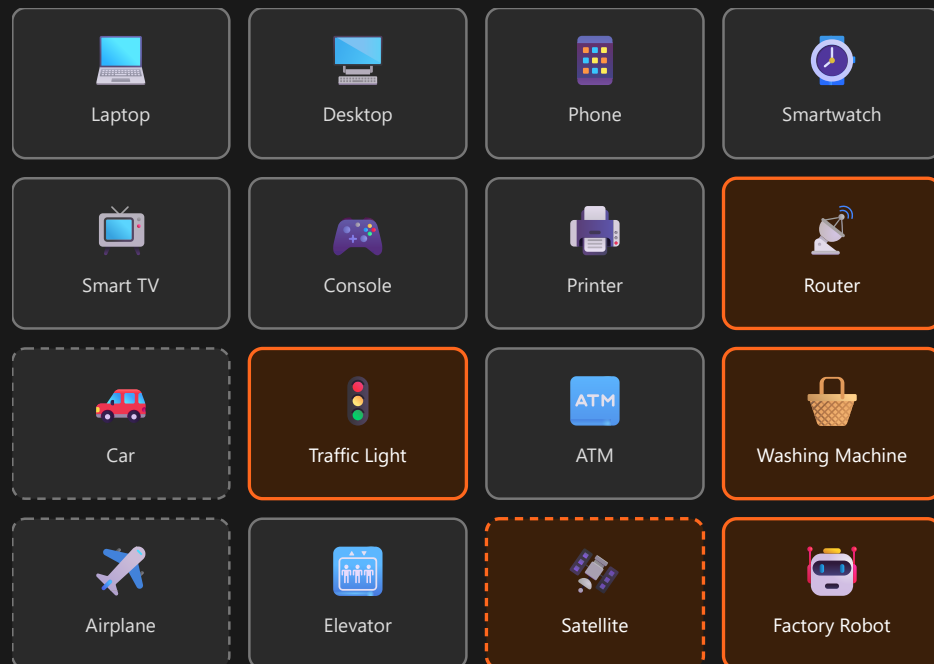
- Latin: distribuere, divide
- 1990s: shortened by users
- Hundreds of variants exist





HOW DO YOU INTERACT WITH THE OS?

- Graphical interface (GUI)
- No display: network or API



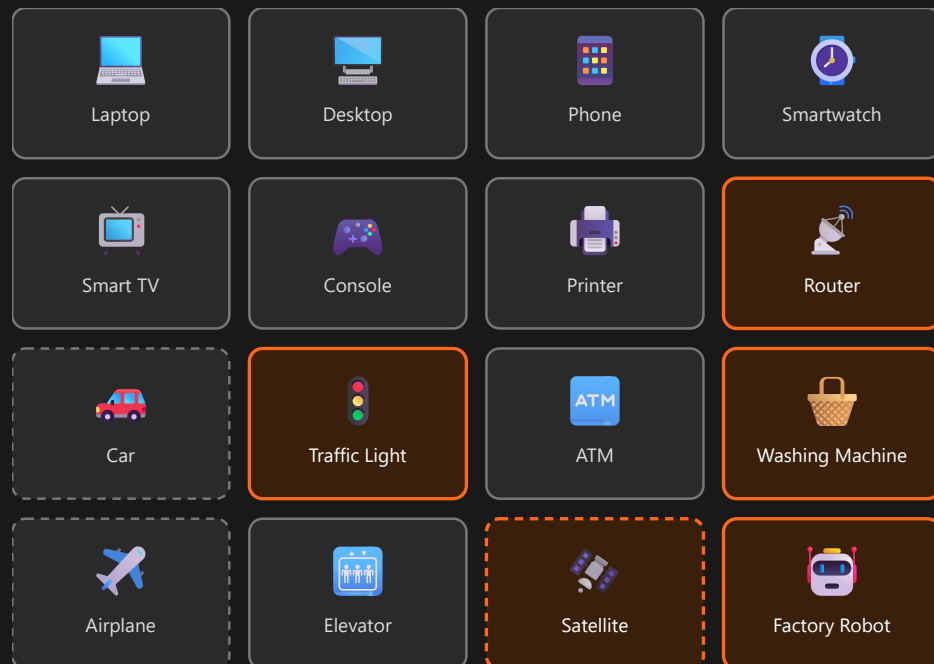
Highlighted: no display · Dashed: not stationary



HOW DO YOU INTERACT WITH THE OS?

- Graphical interface (GUI)
- No display: network or API

→ **Command-line interface** (CLI)



Highlighted: no display · Dashed: not stationary

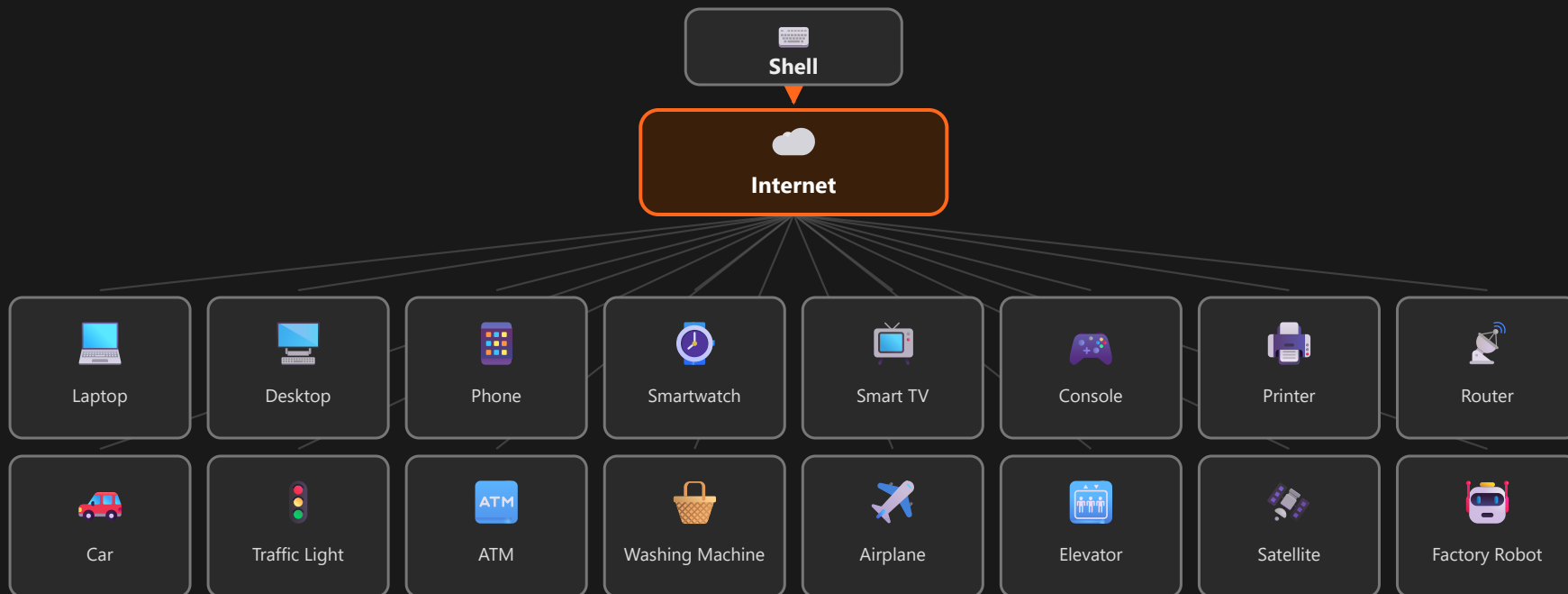


QUESTIONS?



SHELLS: CMD, POWERSHELL & BASH

HOW FAR CAN ONE SHELL REACH?





SHELLS: CMD, POWERSHELL & BASH

- A shell runs what you type
- Windows: cmd.exe, PowerShell
- Linux/macOS/WSL: Bash, zsh
- You'll use all three here

Shell	OS	First Released	Style
cmd.exe	Windows	1981	Simple commands
PowerShell	Windows	2006	Object-based, scriptable
Bash	Linux/macOS/WSL	1989	POSIX, scriptable



SHELLS: REAL EXAMPLES

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.22621.963]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Wikipedia>
```

cmd.exe (Source: [Wikimedia Commons](#))

```
PowerShell
PowerShell 7.5.0
PS C:\Users\John> $PSVersionTable

Name                           Value
----                           -
PSVersion                      7.5.0
BuildVersion                   7.5.0
OS                             Microsoft Windows 10.0.22631
Platform                       Win32NT
PSCoreCLRVersion               [1.0, 2.0, 3.0, 4.0]
PSScriptingEngineVersion       1.1.0.1
WSManStackVersion              3.0

PS C:\Users\John> function Say-Hello {
>> Write-Host "Hello, world!"
>> }
PS C:\Users\John> Say-Hello
Hello, world!
PS C:\Users\John>
```

PowerShell (Source: [Wikimedia Commons](#))

```
fd@fHAL4000:~$ uname -a
Linux fHAL4000 3.4.0-#1 FREEMPT Thu Aug 1 17:06:05 CST 2013 x86_64 x86_64 GNU/Linux
fd@fHAL4000:~$ ls
a.out hello.c
fd@fHAL4000:~$ cat hello.c
#include <stdio.h>

main()
{
    printf("Hello, world!\n");
}
fd@fHAL4000:~$ cc hello.c
fd@fHAL4000:~$ ./a.out
Hello, world!
fd@fHAL4000:~$ cal
      Janvier 2017
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30 31
fd@fHAL4000:~$
```

Bash, via WSL (Source: [Wikimedia Commons](#))



SHELLS: REAL EXAMPLES

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.22621.963]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Wikipedia>
```

cmd.exe (Source: [Wikimedia Commons](#))

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PowerShell
PowerShell 7.5.0
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Name                           Value
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PS C:\Users\John>
```

PowerShell (Source: [Wikimedia Commons](#))

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Linux fHAL4000 3.4.0-#1 FREEMPT Thu Aug 1 17:06:05 CST 2013 x86_64 x86_64 GNU/Linux
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fd@fHAL4000:~$ cc hello.c
fd@fHAL4000:~$ ./a.out
Hello, world!
fd@fHAL4000:~$ cal
      Janvier 2017
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30 31
fd@fHAL4000:~$
```

Bash, via WSL (Source: [Wikimedia Commons](#))



WORD ORIGIN: "SHELL"

1965, MULTICS (MIT)

- Coins the term “shell”
- A program around the OS

1971, UNIX SHELL

- Ken Thompson brings it
- to Unix First Edition



WORD ORIGIN: "GNU"

1983, FOUNDING

- Richard Stallman announces
- the GNU Project, MIT

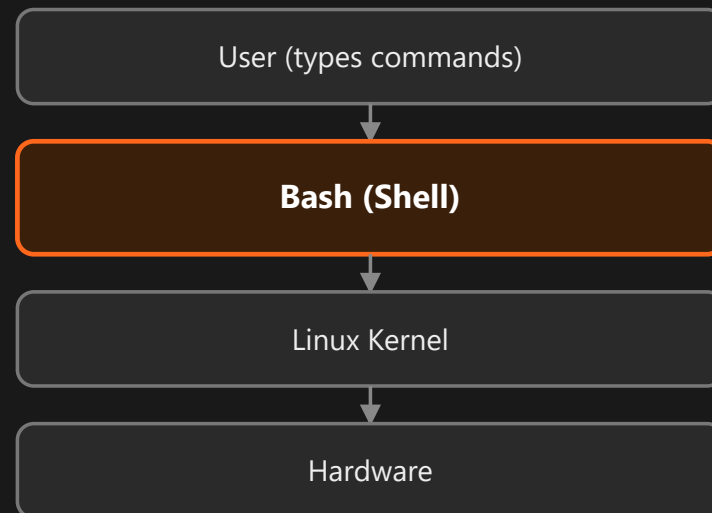
RECURSIVE ACRONYM

- "GNU's Not Unix" – a
- programmer's inside joke



BASH

- Reads the commands you type
- Interprets & runs them
- Passes work to the kernel
- Runs your shell scripts



Bash sits between you and the kernel – every command passes through it.



BASH: REAL EXAMPLES

```
root@localhost ~# ping -q fa.wikipedia.org
PING text.patpa.wikimedia.org (208.80.152.2) 56(84) bytes of data:
64 bytes from 208.80.152.2: icmp_seq=1 ttl=64 time=0.000 ms
--- text.patpa.wikimedia.org ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.000/0.000/0.000/0.000 ms
root@localhost ~# pwd
/root
root@localhost ~# cd /var
root@localhost var# ls -la
total 72
drwxr-xr-x. 18 root root 4096 Jul 30 22:43 .
drwxr-xr-x. 23 root root 4096 Sep 14 20:42 ..
drwxr-xr-x. 2 root root 4096 May 14 00:15 account
drwxr-xr-x. 11 root root 4096 Jul 31 22:25 cache
drwxr-xr-x. 3 root root 4096 May 18 16:03 db
drwxr-xr-x. 3 root root 4096 May 18 16:03 empty
drwxr-xr-x. 2 root root 4096 May 18 16:03 games
drwxr-xr-x. 2 root root 4096 Jun 2 18:39 gsm
drwxr-xr-x. 38 root root 4096 May 18 16:03 lib
drwxr-xr-x. 2 root root 4096 May 18 16:03 local
lrwxrwxrwx. 1 root root 11 May 14 00:12 lock -> ../run/lock
drwxr-xr-x. 14 root root 4096 Sep 14 20:42 log
lrwxrwxrwx. 1 root root 19 Jul 30 22:43 mail -> spool/mail
drwxr-xr-x. 2 root root 4096 May 18 16:03 nis
drwxr-xr-x. 2 root root 4096 May 18 16:03 opt
drwxr-xr-x. 2 root root 4096 May 18 16:03 preserve
drwxr-xr-x. 2 root root 4096 Jul 1 22:11 report
lrwxrwxrwx. 1 root root 6 May 14 00:12 run -> ../run
drwxr-xr-x. 14 root root 4096 May 18 16:03 spool
drwxr-xr-x. 4 root root 4096 Sep 12 23:58 tmp
drwxr-xr-x. 2 root root 4096 May 18 16:03 yp
root@localhost var# yum search wiki
Loaded plugins: langpacks, presto, refresh-packagekit, remove-with-leaves
rpmfusion-free-updates                               | 2.7 kB | 00:00
rpmfusion-free-updates/primary_db                     | 206 kB | 00:04
rpmfusion-nonfree-updates                             | 2.7 kB | 00:00
updates/metalink                                       | 5.9 kB | 00:00
updates                                                 | 4.7 kB | 00:00
updates/primary_db                                     | 73% [=====] | 62 kB/s | 2.6 MB | 00:15 ETA
```

```
[/etc]$ ls
vidarlio
[/home]$ cd ..
[/]$ cd etc
[/etc]$ ls
0,0,10.in-addr.arpa
adduser.conf
adjtime
aliases
alternatives
apm
apt
asterisk
at.deny
bakplungfu
bash.bashrc
bash_completion
bash_completion.d
bind
blkid.tab
blkid.tab.old
calendar
chatscripts
chkrootkit.conf
complete.tcsh
console
console-tools
cron.d
cron.daily
cron.hourly
cron.monthly
crontab
cron.weekly
/etc$
```

osh.cshrc	gshadow	logrotate.d	odbcinst.ini	rpc
osh.login	gtk	lynx.cfg	openoffice	rpc
osh.logout	host.conf	magic	opt	screenc
db.cache	hostname	mailcap	pan.conf	security
debconf.conf	hosts	mailcap.order	pan.d	security
debian_version	hosts.allow	mailname	passwd	services
default	hosts.deny	mail.rc	passwd-	shadow
defoma	hotplug	manpath.config	perl	shadow
deluser.conf	hotplug.d	ndadm	ppp	shells
dchlient.conf	identd.conf	mediasrv	printcap	skel
dchlient-script	identd.key	nime.types	profile	squid
dictionaries-common	inetd.conf	nkinitrd	protocols	ssh
discover.conf	init.d	modprobe.d	python2.3	sudoers
discover.conf-2.6	inittab	modules	raidtab	sysctl.conf
discover.d	inputrc	modules.conf	rc0.d	syslog.conf
dkg	iplungfu	modules.conf.old	rc1.d	terminfo
emacs	issue	modutils	rc2.d	timezone
emacs21	issue.net	notd	rc3.d	ucf.conf
email-addresses	kernel-img.conf	ntab	rc4.d	updatedb.conf
environment	ldap	ntools.conf	rc5.d	vidarlio.net.hosts
exim4	ld.so.cache	Mutttrc	rc6.d	w3m
fdmount.conf	ld.so.conf	mysql	rc7.d	wgetrc
fonts	locale.alias	nanorc	rc8.d	wgetrc
fstab	locale.gen	network	reportbug.conf	wdial.conf#
groff	localtime	networks	resolvconf	wdial.conf
group	logcheck	nsswitch.conf	resolv.conf	wdial.conf
group-	login.defs	ODBCDataSources	resolv.conf	xpilot
gshadow	logrotate.conf	odbc.ini	resolv.conf.pppd-backup	

Bash in GNOME Terminal (Source: [Wikimedia Commons](#))

Same Bash, a different terminal: Xterm (Source: [Wikimedia Commons](#))



BASH: REAL EXAMPLES

```
root@localhost ~# ping -q fa.wikipedia.org
PING text.patpa.wikimedia.org (208.80.152.2) 56(84) bytes of data:
64 bytes from 208.80.152.2: icmp_seq=1 ttl=64 time=0.000 ms
--- text.patpa.wikimedia.org ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.000/0.000/0.000/0.000 ms
root@localhost ~# pwd
/root
root@localhost ~# cd /var
root@localhost /var# ls -la
total 72
drwxr-xr-x. 18 root root 4096 Jul 30 22:43 .
drwxr-xr-x. 23 root root 4096 Sep 14 20:42 ..
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updates/metalink | 5.9 kB | 00:00
updates | 4.7 kB | 00:00
updates/primary_db | 73% [=====] | 62 kB/s | 2.6 MB | 00:15 ETA
```

Bash in GNOME Terminal (Source: [Wikimedia Commons](#))

```
[/home]$ ls
vidarlo
[/home]$ cd ..
[/]$ cd etc
[/etc]$ ls
0,0,10.in-addr.arpa osh.cshrc gshadow logrotate.d odboinst.ini rat
adduser.conf osh.login gtk host.conf lynx.cfg openoffice
adtime osh.logout host.conf magic opt openoffice
aliases db.cache hostname mailcap pam.conf pam.d security
alternatives debconf.conf hosts.allow mailname passwd services
apm apt default hosts.deny mail.rc manpath.config perl shadow
asterisk deluser.conf hotplug hotplug.d mdadm ppp printcap shells
at.deny baliplungfu dcclient.conf identd.conf mediawiki profile squid
bash.bashrc dcclient-script identd.key mime.types protocols ssh
bash_completion dictionaries-common inetd.conf nkinitrd python2.3
bash_completion.d discover.conf init.d inittab modules sudoers
bind discover.conf-2.6 inputrc ipkginfo modules.conf rc0.d sysctl.conf
blkid.tab discover.d ipkginfo modules.conf,old rc1.d syslog.conf
blkid.tab,old dkg issue modules.conf,old rc2.d terminfo
calendar emacs issue.net notd rc3.d timezone
chatscripts chkrootkit.conf email-addresses kernel-img.conf rc4.d ucf.conf
complete,tcsh environment ldap mtools.conf rc5.d updatedb.conf
console exim4 ld.so.cache muttrc rc6.d vidarlo.net.hosts
console-tools fdmout.conf ld.so.conf mysql rc7.d w3m
cron.d fonts locale.alias nanorc rc8.d wgetrc
cron.daily fstab localtime network reportbug.conf wvdial.conf#
cron.hourly groff logcheck networks resolvconf wvdial.conf
cron.monthly group logcheck nsswitch.conf resolv.conf wvdial.conf
crontab group login.defs ODBCDataSources resolv.conf wvdial.conf
cron.weekly group- gshadow logrotate.conf resolv.conf.pppd-backup
[/etc]$
```

Same Bash, a different terminal: Xterm (Source: [Wikimedia Commons](#))

The terminal is just a window – Bash is the same shell either way. Don't confuse "terminal emulator" with "shell."



WORD ORIGIN: "BASH"

1979, THE BOURNE SHELL

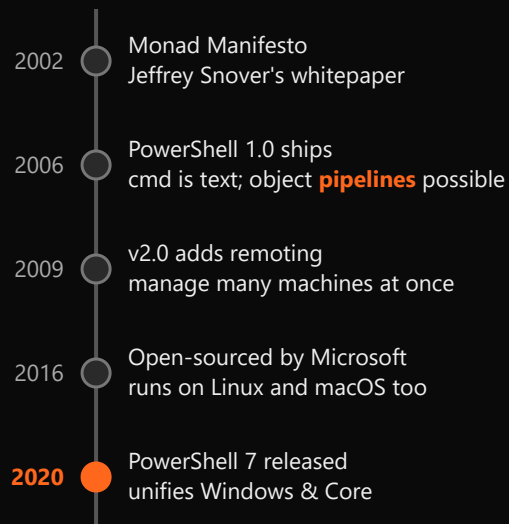
- Stephen Bourne writes "sh"
- at Bell Labs

1989, BASH

- Brian Fox, GNU Project –
- "Bourne-Again SHell," a pun



WHY POWERSHELL, NOT CMD, FOR ADMINS?



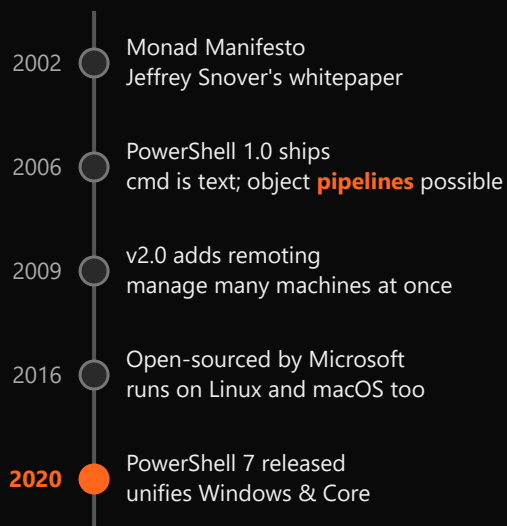
```
$ Get-Process # running processes
$ Get-Service # Windows services
$ Get-ChildItem # list files (ls)
$ Get-Content # read a file
$ Get-NetIPAddress # show IP config
$ Get-EventLog # view event logs
$ Get-Help # cmdlet docs
$ Get-Member # object properties
$ Set-ExecutionPolicy # allow/block scripts
$ Get-History # past commands
```

Copy

PowerShell cmdlet reference (Source: [Microsoft Learn](#))



WHY POWERSHELL, NOT CMD, FOR ADMINS?



→ **Start your computer and open PowerShell**

```
$ Get-Process # running processes
$ Get-Service # Windows services
$ Get-ChildItem # list files (ls)
$ Get-Content # read a file
$ Get-NetIPAddress # show IP config
$ Get-EventLog # view event logs
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$ Get-History # past commands
```

PowerShell cmdlet reference (Source: [Microsoft Learn](#))



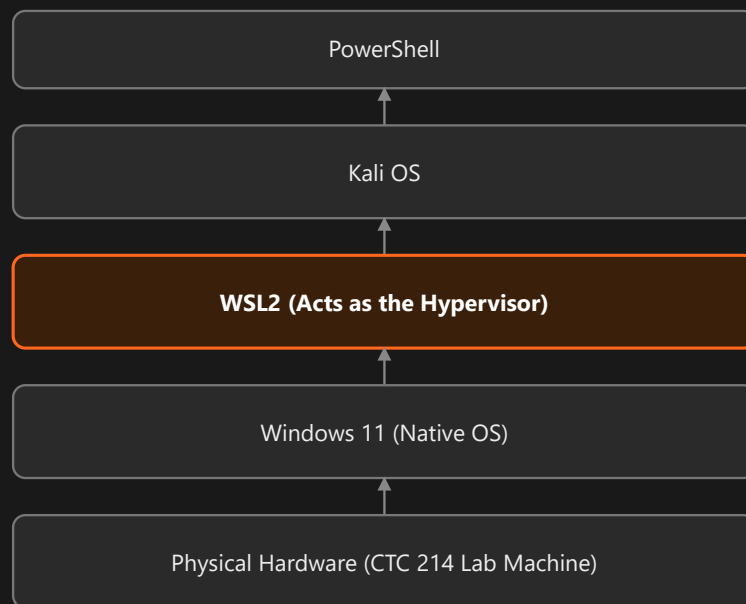
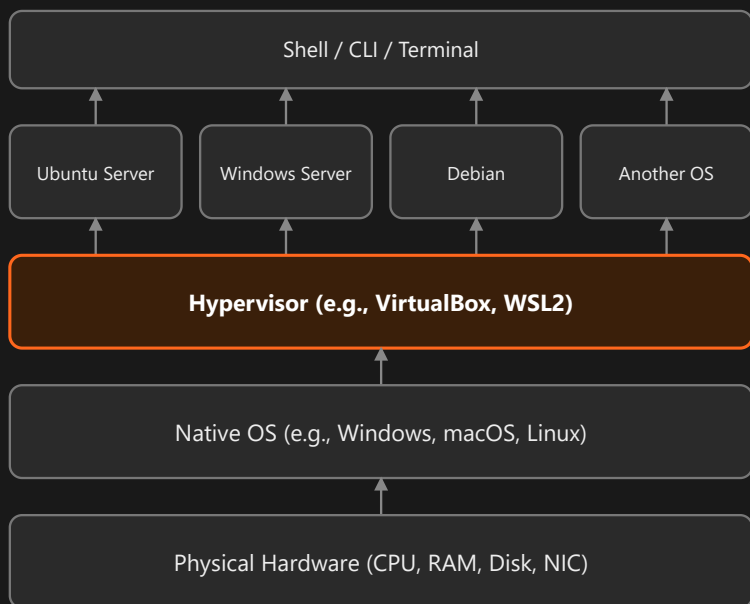
QUESTIONS?



HANDS-ON: INSTALL LINUX

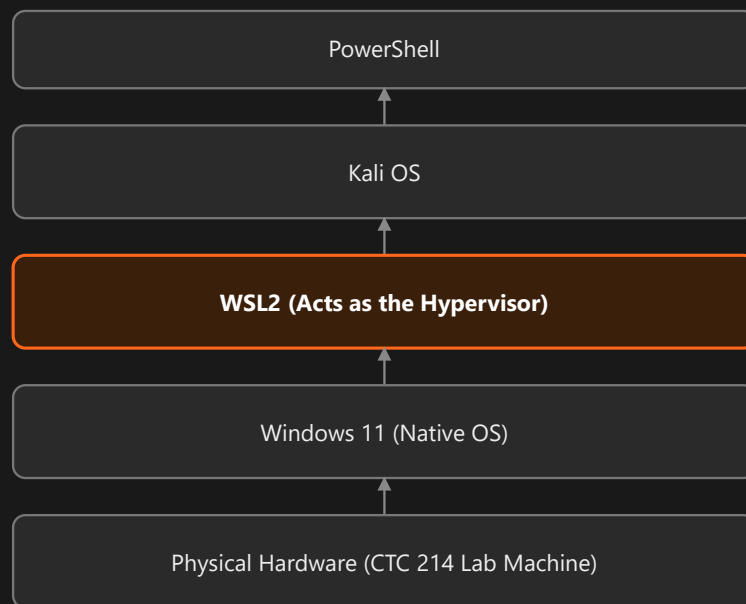
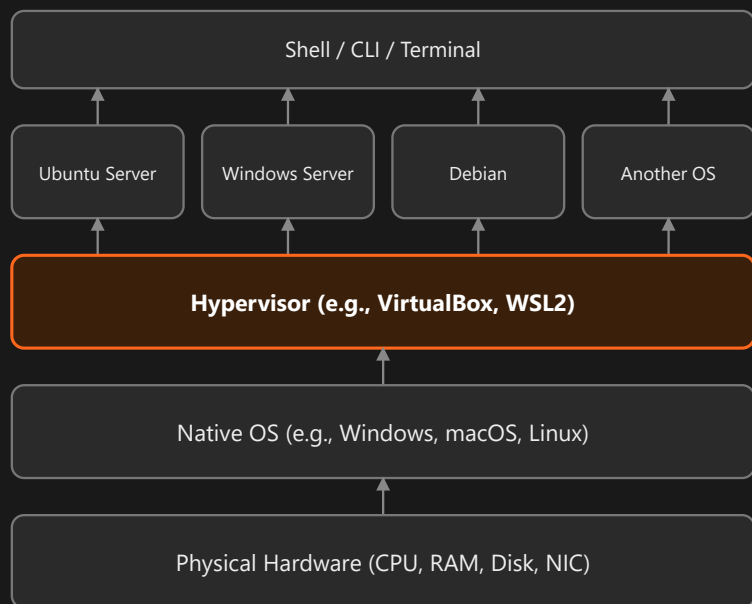


WHAT IS VIRTUALIZATION?





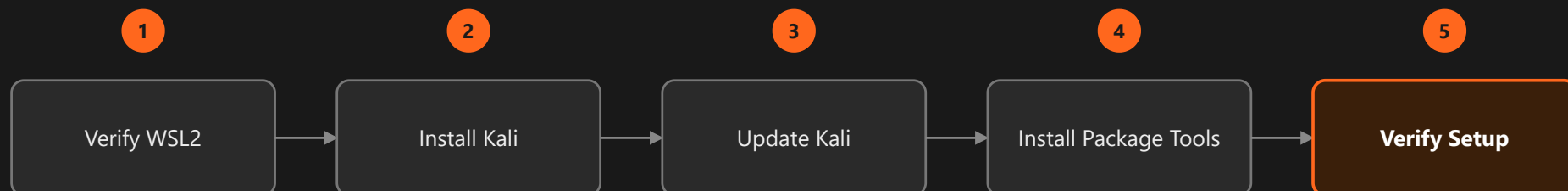
WHAT IS VIRTUALIZATION?



⚠ Don't assume the guest OS "knows" it's virtualized. It thinks the hardware is **real**.



KALI INSTALL: MAIN STEPS





WHY KALI LINUX?

- Debian-based: same apt
- Ships 600+ security tools
- Industry-standard, widely used
- Officially supports WSL2
- No admin rights needed here



Kali Linux logo (Source: [Wikimedia Commons](#))



OPEN POWERSHELL

- Launch PowerShell normally
- Do NOT run as Administrator
- WSL2 is already enabled

```
$ wsl --status  
$ wsl --version
```

[Copy](#)



CHECK FOR KALI

- List installed distributions
- Fresh account: none listed
- Confirms WSL2, not WSL1

```
$ wsl --list --verbose
```

[Copy](#)



INSTALL KALI

- `wsl --install -d kali-linux`
- Blocked? Add `--web-download`
- Create Kali username/password
- Not your University password
- Set Kali as your default

```
$ wsl --list --online
$ wsl --set-default-version 2
$ wsl --install -d kali-linux
$ wsl --list --verbose
$ wsl --set-default kali-linux
```

[Copy](#)



INSTALL KALI

- `wsl --install -d kali-linux`
- Blocked? Add `--web-download`
- Create Kali username/password
- Not your University password
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```
$ wsl --list --online
$ wsl --set-default-version 2
$ wsl --install -d kali-linux
$ wsl --list --verbose
$ wsl --set-default kali-linux
```

Copy

⚠️ Kali login, sudo, and VNC passwords are all separate from your University password. **Never** reuse it here.



START KALI

- Appears in the Start menu
- Or start it from PowerShell
- Opens a Linux shell prompt

```
$ wsl -d kali-linux
```

[Copy](#)



UPDATE KALI FIRST

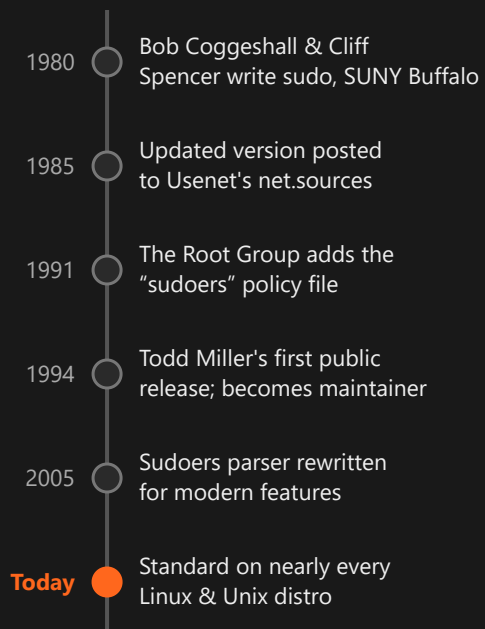
- Run these inside Kali
- full-upgrade, not upgrade
- Kali is a rolling distro

```
$ sudo apt update  
$ sudo apt full-upgrade -y
```

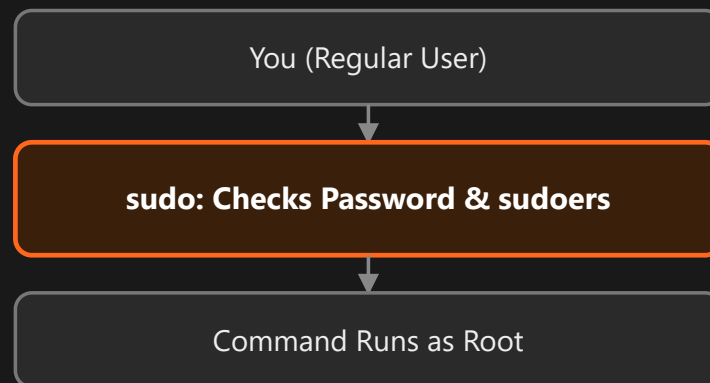
[Copy](#)



WORD ORIGIN: "SUDO"

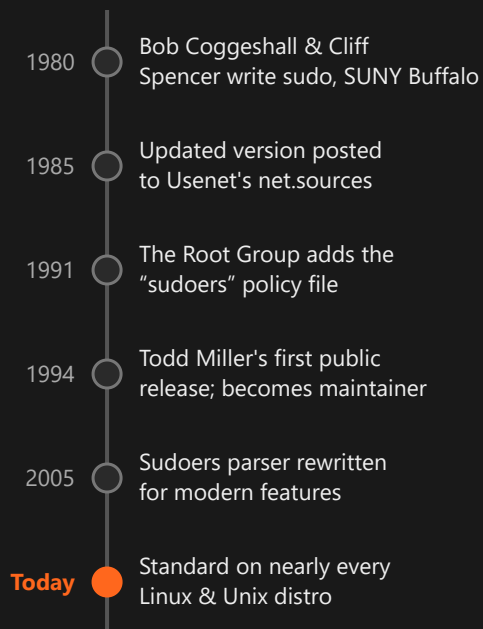


WHAT DOES **SUDO** DO?

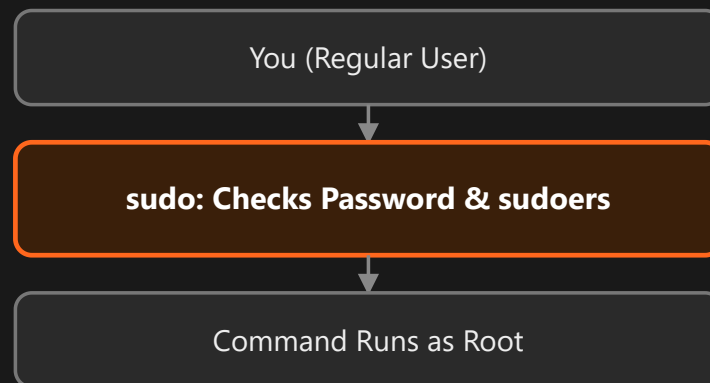




WORD ORIGIN: "SUDO"



WHAT DOES SUDO DO?



⚠ Privileges return to normal the moment the command finishes. sudo doesn't make you root **permanently**.



INSTALL WIN-KEX & TOOLS

- Installs the GUI + toolset
- kali-linux-default = full set
- Ready-to-use security tools
- Large download, start early

```
$ sudo apt install -y git #
$ sudo apt install -y curl #
$ sudo apt install -y nmap #
$ sudo apt install -y nikto #
$ sudo apt install -y hydra #
$ sudo apt install -y sqlmap #
$ sudo apt install -y john #
$ sudo apt install -y gobuster #
$ sudo apt install -y kali-win-kex #
$ sudo apt install -y kali-linux-default #
```

Full package list (Source: kali.org/tools)



INSTALL WIN-KEX & TOOLS

- Installs the GUI + toolset
- kali-linux-default = full set
- Ready-to-use security tools
- Large download, start early

```
$ sudo apt install -y git #
$ sudo apt install -y curl #
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$ sudo apt install -y sqlmap #
$ sudo apt install -y john #
$ sudo apt install -y gobuster #
$ sudo apt install -y kali-win-kex #
$ sudo apt install -y kali-linux-default #
```

Full package list (Source: kali.org/tools)

⚠ This step can take a long time and a lot of disk space. Start it during a lecture day, well **before** you need the GUI.



FILES & WINDOWS ACCESS

- Local to this one computer
- Home: /home/your-username
- Windows C: at /mnt/c
- sudo is not Windows admin

```
$ ls /home/$USER  
$ ls /mnt/c/Users
```

[Copy](#)



RESETTING KALI (NOT NOW)

- Deletes everything, careful
- Not routine troubleshooting
- Reinstall fresh afterward

```
$ wsl --unregister kali-linux  
$ wsl --install -d kali-linux
```

Copy



RESETTING KALI (NOT NOW)

- Deletes everything, careful
- Not routine troubleshooting
- Reinstall fresh afterward

🔔 This permanently deletes your entire Kali install, including your Linux home directory and any coursework stored only there.

```
$ wsl --unregister kali-linux  
$ wsl --install -d kali-linux
```

Copy



CHOOSING A USERNAME

- Lowercase, digits, hyphens
- Starts with a letter
- 32 characters or fewer
- e.g., jsmith, jane.smith
- Role-based: dbadmin, websvc

```
$ whoami           # your own username
$ id               # UID and groups
$ passwd           # change your password
$ sudo usermod -l new old # rename a username
```



CHOOSING A USERNAME

- Lowercase, digits, hyphens
- Starts with a letter
- 32 characters or fewer
- e.g., jsmith, jane.smith
- Role-based: dbadmin, websvc

```
$ whoami                # your own username
$ id                    # UID and groups
$ passwd                # change your password
$ sudo usermod -l new old # rename a username
```

Break this rule and Kali will flatly reject the username. It's not just a suggestion.



CHOOSING A PASSWORD

- Length beats complexity
- 15+ characters, a passphrase
- Unique per system or account
- Use a password manager
- No forced periodic changes

```
$ passwd # change your own pa
$ sudo passwd jsmith # change another use
$ passwd -S # check password sta
$ chage -l jsmith # view expiry info
$ sudo chage -M 30 jsmith # passwor
$ sudo chage -E 2026-11-21 jsmith # account
```

Password & account-aging commands (Source: [NIST SP 800-63B](#))



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$ sudo chage -M 30 jsmith # passwor
$ sudo chage -E 2026-11-21 jsmith # account
```

Password & account-aging commands (Source: [NIST SP 800-63B](#))

Forced periodic password changes make passwords weaker, not stronger – don't do it.



VERIFY (CLI)

- Confirm OS & kernel
- Confirm Windows files visible
- Update the system

```
$ uname -a  
$ ls /mnt/c/Users  
$ sudo apt update && sudo apt full-upgrade -
```

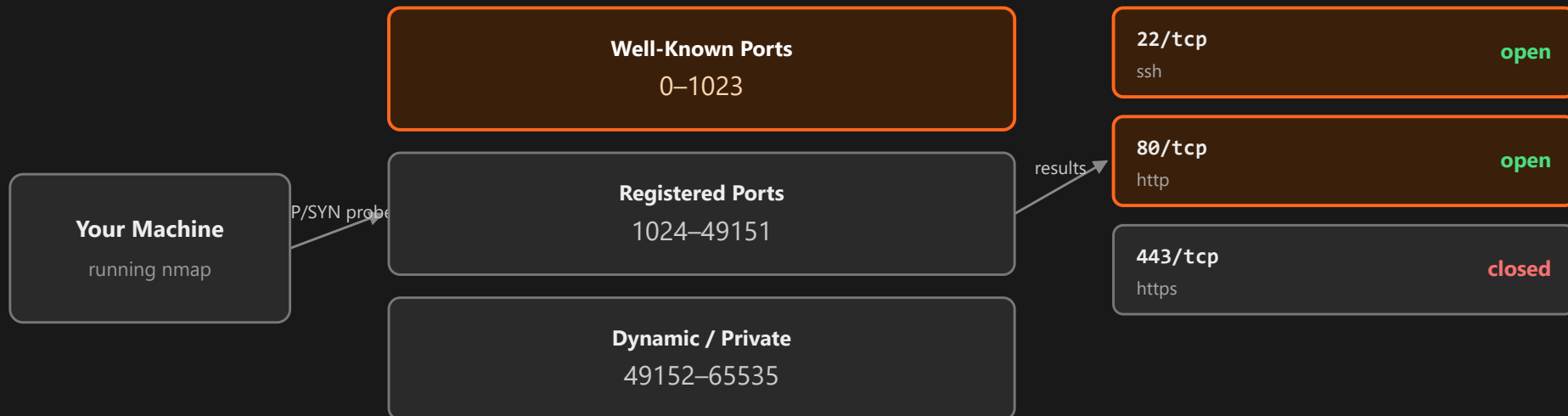
Copy



WHAT IS NMAP?

Target Host: Operating System

Scan Results





WHERE DO YOU FIND NMAP DOCS?

- `man nmap`: full reference
- `nmap -h`: quick help
- Official: Nmap Reference Guide

```
$ man nmap          # full manual page
$ nmap -h           # quick help summary
$ nmap --version    # check installed version
```

Copy

Nmap Reference Guide (Source: nmap.org)



WHAT ARE SOME SAMPLE NMAP CALLS?

- -sn: host discovery only
- -p 22,80: specific ports
- -A: OS & version detect

```
$ nmap scanme.nmap.org      # default scan
$ nmap -sn scanme.nmap.org  # host discovery only
$ nmap -p 22,80 scanme.nmap.org # specific ports
$ nmap -A scanme.nmap.org    # OS & version detect
```

Sample calls against Nmap's own test target (Source: scanme.nmap.org)



WHAT DOES THE OUTPUT MEAN?

- Real scan: scanme.nmap.org
- 4 open ports, OS guessed
- 31337: a classic 'leet' port

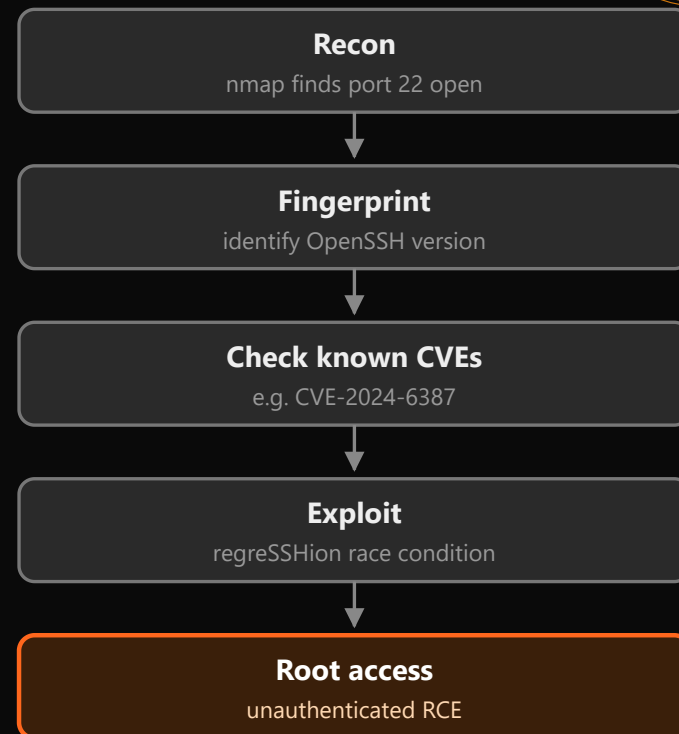
```
Nmap scan report for scanme.nmap.org (45.33.
Host is up (0.024s latency)           # host res
Not shown: 996 filtered ports        # most por
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9929/tcp   open  nping-echo
31337/tcp open  Elite
Running: Linux 4.X|5.X               # OS gues
Network Distance: 15 hops            # hops t
```

Live scan output, run by the instructor against
scanme.nmap.org (Source: sudo nmap -O scanme.nmap.org,
Aug 2026)



WHY DOES AN OPEN PORT MATTER?

- Each port = attack surface
- Old software often unpatched
- One flaw can mean full access



CVE-2024-6387 "regreSSHion," OpenSSH unauthenticated RCE
(Source: [Qualys](#), Jul 2024)



WHERE CAN YOU LEGALLY PRACTICE?

- scanme.nmap.org: official
- TryHackMe, HackTheBox: labs
- Metasploitable2: local VM

```
$ nmap scanme.nmap.org      # Nmap's own te
$ nmap 10.10.x.x            # TryHackMe VPN
$ nmap 10.10.x.x            # HackTheBox VP
$ nmap 192.168.56.101       # local Metasp1
```

Public scan targets (Source: scanme.nmap.org)



ONLY SCAN WHAT YOU'RE AUTHORIZED TO SCAN

- scanme.nmap.org, THM, HTB
- Metasploitable2: your own VM
- Nothing else, ever

“The best way to avoid controversy when using Nmap is to always secure written authorization from the target network representatives before initiating any scanning.”

Nmap, Legal Issues (Source: nmap.org)



ONLY SCAN WHAT YOU'RE AUTHORIZED TO SCAN

- scanme.nmap.org, THM, HTB
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- Nothing else, ever

“The best way to avoid controversy when using Nmap is to always secure written authorization from the target network representatives before initiating any scanning.”

Nmap, Legal Issues (Source: nmap.org)

🚨 These public sandboxes are the **only** systems you may scan. Never point nmap at any other server, on or off campus, without written authorization.



HOW DO YOU SCAN JUST ONE PORT?

- Faster than a full scan



HOW DO YOU SCAN JUST ONE PORT?

- Faster than a full scan

```
$ nmap -p 80 scanme.nmap.org # scan only po
```



HOW DO YOU DETECT THE OS?

- Needs root / sudo privileges



HOW DO YOU DETECT THE OS?

- Needs root / sudo privileges

```
$ sudo nmap -O scanme.nmap.org # guess the
```



OS DETECTION: REAL OUTPUT

- Same 4 ports, still open
- OS guess: Linux 4.15–5.19
- 16 hops to the target

```
$ sudo nmap -O scanme.nmap.org
```

PORT	STATE	SERVICE
------	-------	---------

22/tcp	open	ssh
--------	------	-----

80/tcp	open	http
--------	------	------

9929/tcp	open	nping-echo
----------	------	------------

31337/tcp	open	Elite
-----------	------	-------

Device type: general purpose	# broad
------------------------------	---------

Running: Linux 4.X 5.X	# OS fa
------------------------	---------

OS details: Linux 4.15 - 5.19	# narro
-------------------------------	---------

Network Distance: 16 hops

Live scan output, run by the instructor against
scanme.nmap.org (Source: sudo nmap -O scanme.nmap.org,
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31337/tcp open  Elite
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Running: Linux 4.X|5.X                 # OS fa
OS details: Linux 4.15 - 5.19          # narro
Network Distance: 16 hops
```

Live scan output, run by the instructor against
scanme.nmap.org (Source: `sudo nmap -O scanme.nmap.org`,
Aug 2026)

● **Red Team:** the OS/version guess narrows which known exploits are worth trying against this target.

● **Blue Team:** the same fingerprint tells defenders what to patch, and whether to suppress it from scanners.



HEADS UP: OS GUESSES AREN'T CERTAIN

⚠ Nmap warns that OS-scan results may be **unreliable** when it can't find at least one open and one closed port on the target. Treat the guess as a best estimate, not a certainty.



HOW DO YOU SCAN ALL 65,535 PORTS?

- Slower, but thorough



HOW DO YOU SCAN ALL 65,535 PORTS?

- Slower, but thorough

```
$ nmap -p- scanme.nmap.org # scan every TCP
```



HOW DO YOU LOOK UP DOMAIN INFO?

- Public registration record
- Needs the base domain



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```
$ whois nmap.org # domain registration info
```



HOW DO YOU LOOK UP DOMAIN INFO?

- Public registration record
- Needs the base domain

```
$ whois nmap.org # domain registration info
```

● **Red Team:** registrant, registrar & name-server info feeds phishing pretexts and social engineering.

● **Blue Team:** confirms your own domain hasn't expired or been hijacked; profiles lookalike phishing domains.



HEADS UP: WHOIS NEEDS THE BASE DOMAIN

⚠ WHOIS looks up the registered **domain**, not a subdomain. ``whois scanme.nmap.org`` fails with "Malformed request" on some registries; use the base domain instead.



HOW DO YOU LOOK UP DNS RECORDS?

- Resolves hostname to IP



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```
$ dig scanme.nmap.org # DNS lookup
```

[Copy](#)



HOW DO YOU LOOK UP DNS RECORDS?

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$ dig scanme.nmap.org # DNS lookup
```

[Copy](#)

● **Red Team:** maps mail servers, subdomains & DNS infrastructure to find attack surface.

● **Blue Team:** verifies DNS resolves correctly and that SPF/DKIM/DMARC anti-phishing records are published.



QUESTIONS?



UNIVERSITY DOMAIN: PLAN VM



SCENARIO: UNIVERSITY DOMAIN

You're the new sysadmin at Pacific University. Every setup choice below follows this one VM for the rest of the course.



UNIVERSITY DOMAIN, BY THE NUMBERS

6,852

Students

890

Faculty

1,228

Staff

3

Campuses

9

Schools & Colleges

30,000 sq ft

Medical Center, Sacramento

University of the Pacific, Fall 2025/2024 Fast Facts & Dugoni School newsroom (Source: [pacific.edu](https://www.pacific.edu))



VM NAME

- Pattern: role-env-number
- Lowercase, hyphens only
- Must be unique on network

UNIVERSITY DOMAIN

- Registrar's web server
- reg-web01
- Instantly IDs owner & role



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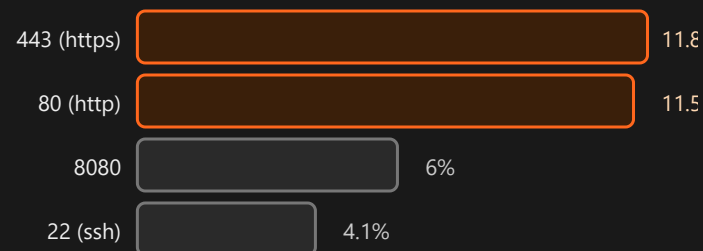
A vague name like "server1" becomes unmanageable once you have 50 VMs.



WHY A WEB SERVER FIRST?

- Top ports: 80/443 (web)
- Most exposed service, by far
- That's why reg-web01 is first

Most-Exposed Internet Ports



Share of devices exposing each port, Shodan scan data
(Source: [SecureBin](#), [Open Port Security Risks](#))



USERNAME

- Lowercase, digits, hyphens
- Must start with a letter

UNIVERSITY DOMAIN

- Registrar staff need access
- reg-jsmith
- Ties account to department



USERNAME

- Lowercase, digits, hyphens
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UNIVERSITY DOMAIN

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- reg-jsmith
- Ties account to department

A generic "admin" account can't be traced back to a person when something goes wrong.

PASSWORD

- Length beats complexity
- 15+ characters if no MFA
- VM holds student records
- FERPA protects that data

Time it takes a hacker to brute force your password in 2026

Hardware: 16 x RTX 5090 | Password hash: bcrypt (10)

Number of Characters	Numbers Only	Lowercase Letters	Upper and Lowercase Letters	Numbers, Upper and Lowercase Letters	Numbers, Upper and Lowercase Letters, Symbols
4	Instantly	Instantly	Instantly	Instantly	Instantly
5	Instantly	Instantly	46 mins	2 hours	3 hours
6	Instantly	37 mins	2 days	5 days	1 week
7	Instantly	16 hours	3 months	10 months	2 years
8	Instantly	2 weeks	12 years	50 years	132 years
9	2 hours	1 year	636 years	3k years	9k years
10	20 hours	32 years	33k years	191k years	645k years
11	1 week	839 years	1m years	11m years	45m years
12	3 months	21k years	89m years	737m years	3bn years
13	2 years	567k years	4bn years	45bn years	221bn years
14	23 years	14m years	241bn years	2tn years	15tn years
15	229 years	383m years	12tn years	175tn years	1qd years
16	2k years	9bn years	653tn years	10qd years	75qd years
17	22k years	259bn years	33qd years	675qd years	5qn years
18	228k years	6tn years	1qn years	41qn years	372qn years



Hive Systems

Read more and download at
hivesystems.com/password

Password crack-time by length & character set (Source: [Hive Systems, 2026 Password Table](#))

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Hive Systems

Read more and download at hivesystems.com/password

Password crack-time by length & character set (Source: [Hive Systems, 2026 Password Table](#))

A breach here isn't just an IT headache – it's a federal FERPA violation.



WHICH OS DO YOU RECOMMEND?

- Ubuntu
- Debian
- Kali Linux
- AlmaLinux
- openSUSE

```
$ wsl --install -d Ubuntu-24.04 # LT
$ wsl --install -d Debian # ve
$ wsl --install -d kali-linux # 60
$ wsl --install -d AlmaLinux-9 # RH
$ wsl --install -d openSUSE-Leap-15.6 # en
```



WHICH OS DO YOU RECOMMEND?

- Ubuntu
- Debian
- Kali Linux
- AlmaLinux
- openSUSE

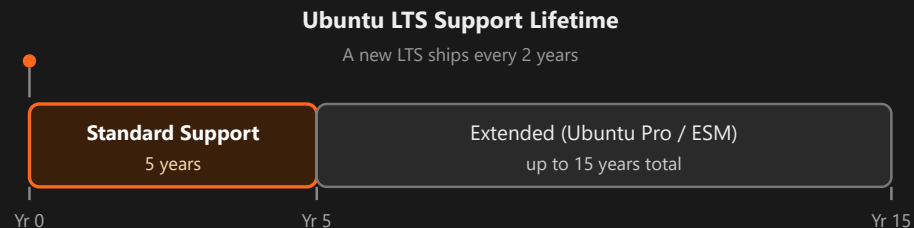
→ **Pick one and install it**

```
$ wsl --install -d Ubuntu-24.04 # LT
$ wsl --install -d Debian # ve
$ wsl --install -d kali-linux # 60
$ wsl --install -d AlmaLinux-9 # RH
$ wsl --install -d openSUSE-Leap-15.6 # en
```



WHICH VM (OS)

- Ubuntu Server LTS
- Matches this course's build
- Records systems run for years
- Matches campus IT standards

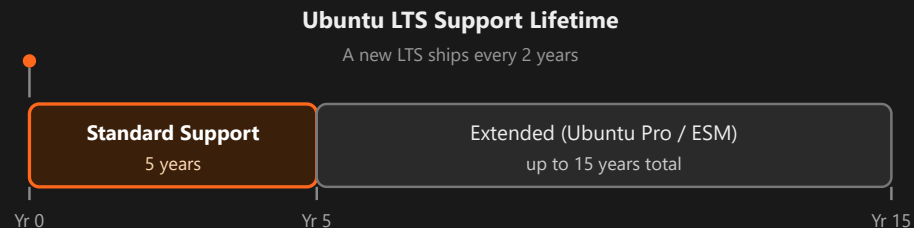


Ubuntu LTS support timeline (Source: [Ubuntu, Release Cycle](#))



WHICH VM (OS)

- Ubuntu Server LTS
- Matches this course's build
- Records systems run for years
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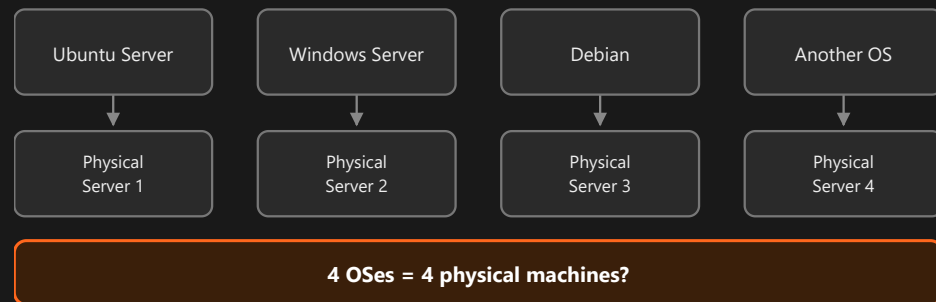
Ubuntu LTS support timeline (Source: [Ubuntu, Release Cycle](#))

A fast-release distro here means re-certifying the whole system every 6 months – university IT can't move that fast.



HOW DO WE SCALE OS ACROSS A UNIVERSITY?

- One OS per server, today
- Servers sit mostly idle
- Buying more is expensive

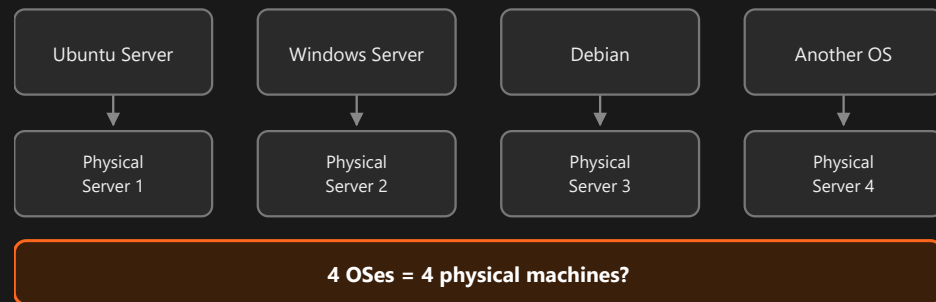




HOW DO WE SCALE OS ACROSS A UNIVERSITY?

- One OS per server, today
- Servers sit mostly idle
- Buying more is expensive

→ Does each OS need its own hardware?





PRACTICE: LOG IN & GO HOME (UBUNTU)

- Starts in your home dir
- Only you can write here
- ~ is shorthand for it

```
$ wsl -d Ubuntu-24.04
$ pwd
/home/yourname
$ whoami
yourname
```

[Copy](#)



PRACTICE: LOG IN & GO HOME (UBUNTU)

- Starts in your home dir
- Only you can write here
- ~ is shorthand for it

⚠ Every user gets exactly one **home** directory (``/home/<username>``). It's the only place you're guaranteed write access without ``sudo``, which is why every exercise in this course starts there instead of somewhere like ``/etc`` or ``/var``.

```
$ wsl -d Ubuntu-24.04
$ pwd
/home/yourname
$ whoami
yourname
```

[Copy](#)



PRACTICE: CREATE THE CATALOG PAGE (UBUNTU)

- Create index.html locally
- Book list = the response body

```
$ mkdir library && cd library
$ cat > index.html << 'EOF'
<h1>Pacific Library Catalog</h1>
<ul><li>Introduction to System
Admin and Security</li></ul>
EOF
```

[Copy](#)



PRACTICE: SERVE LIB-WEB01 (UBUNTU)

- Serves the current directory
- Listens on port 8080

```
$ python3 -m http.server 8080
```

```
Serving HTTP on 0.0.0.0 port 8080 ...
```

[Copy](#)



PRACTICE: FIND LIB-WEB01'S IP (UBUNTU)

- Every distro gets its own IP
- Yours will differ, that's OK

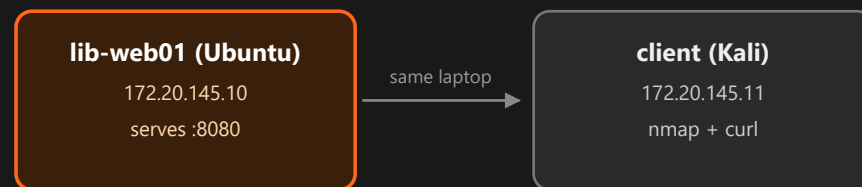
```
$ hostname -I  
172.20.145.10
```

[Copy](#)



PRACTICE: TWO MACHINES, ONE LAPTOP

- lib-web01 hosts the catalog
- client tests the connection



Ubuntu + Kali, one physical machine



PRACTICE: CONFIRM FROM THE CLIENT (KALI)

- nmap confirms the port is open
- curl fetches the real page

```
$ nmap -p 8080 172.20.145.10  
$ curl http://172.20.145.10:8080
```

[Copy](#)



PRACTICE: CONFIRM FROM THE CLIENT (KALI)

- nmap confirms the port is open
- curl fetches the real page

```
$ nmap -p 8080 172.20.145.10  
$ curl http://172.20.145.10:8080
```

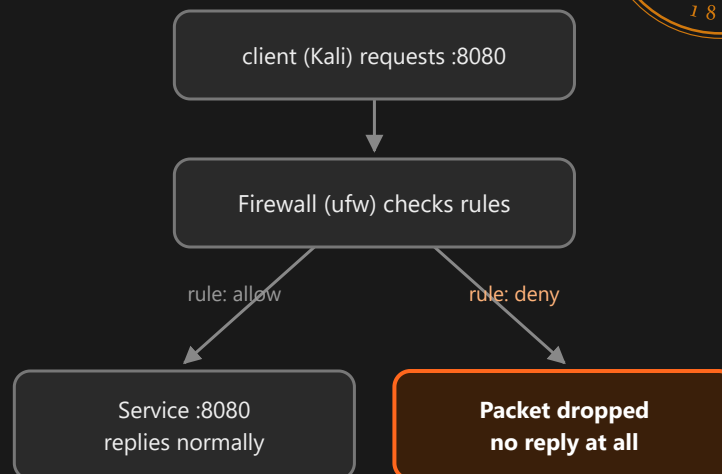
Copy

- **Blue Team:** confirming a port from another machine is exactly how you'd verify your own service is actually reachable, not just "should be" reachable on paper.



HOW DOES A FIREWALL BLOCK A PORT?

- Firewall = a rule checkpoint
- Allow: packet passes through
- Deny: packet just vanishes

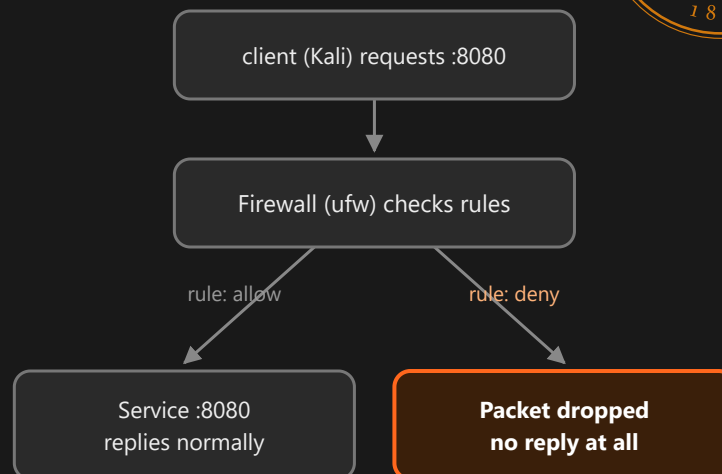


Port 8080 is registered with IANA as "http-alt" (HTTP Alternate) (Source: [IANA, Service Name and Transport Protocol Port Number Registry](#))



HOW DOES A FIREWALL BLOCK A PORT?

- Firewall = a rule checkpoint
- Allow: packet passes through
- Deny: packet just vanishes



● **Red Team:** at a university, attackers scan every public IP for forgotten open ports, like an old admin panel left listening on 8080.

● **Blue Team:** campus IT firewalls every server down to only the ports a service actually needs. 8080 is a well-known dev/alt-HTTP port and should never be public.

Port 8080 is registered with IANA as "http-alt" (HTTP Alternate) (Source: [IANA, Service Name and Transport Protocol Port Number Registry](#))



PRACTICE: BLOCK THE PORT (UBUNTU)

- ufw = simple firewall frontend
- Deny inbound on port 8080
- Allow just one trusted subnet

```
$ sudo apt install -y ufw  
$ sudo ufw enable  
$ sudo ufw deny 8080/tcp  
$ sudo ufw allow from 172.20.145.0/24 to any
```



PRACTICE: TRY AGAIN FROM THE CLIENT (KALI)

- Port now blocked, not open
- filtered = silently dropped

```
$ nmap -p 8080 172.20.145.10  
$ curl http://172.20.145.10:8080
```

[Copy](#)



PRACTICE: TRY AGAIN FROM THE CLIENT (KALI)

- Port now blocked, not open
- filtered = silently dropped

⚠ nmap now reports port 8080 as **filtered**, not ``closed``. ``closed`` means a host replied "nothing's listening here," while ``filtered`` means nothing replied at all, because ufw silently dropped the packet. ``curl`` just hangs until it times out.

```
$ nmap -p 8080 172.20.145.10  
$ curl http://172.20.145.10:8080
```

[Copy](#)



WHAT IS IANA?

- IANA = the global registry
- Assigns port numbers & names
- Free to search, no login

Not sure what a port is for? Look it up:

<https://www.iana.org/assignments/service-names-port-numbers?search=8080>



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<https://www.iana.org/assignments/protocol-parameters/protocol-parameters.html#search=8080>

Try it: look up 3 ports (not 80, 443, or 8080) that you think are useful to know.



PRACTICE: REAL DOMAIN VS. YOUR SERVER (KALI)

- nmap.org: real WHOIS + DNS
- lib-web01: private, unlisted

```
$ whois nmap.org  
$ dig scanme.nmap.org
```

[Copy](#)

```
$ whois 172.20.145.10  
$ dig -x 172.20.145.10
```



FROM PRACTICE TO CONCEPT

You just ran **two** operating systems, Ubuntu and Kali, on one physical laptop, at the same time.



FROM PRACTICE TO CONCEPT

You just ran **two** operating systems, Ubuntu and Kali, on one physical laptop, at the same time.

How can we have multiple OS on one physical machine?



QUESTIONS?



WHAT IS VIRTUALIZATION



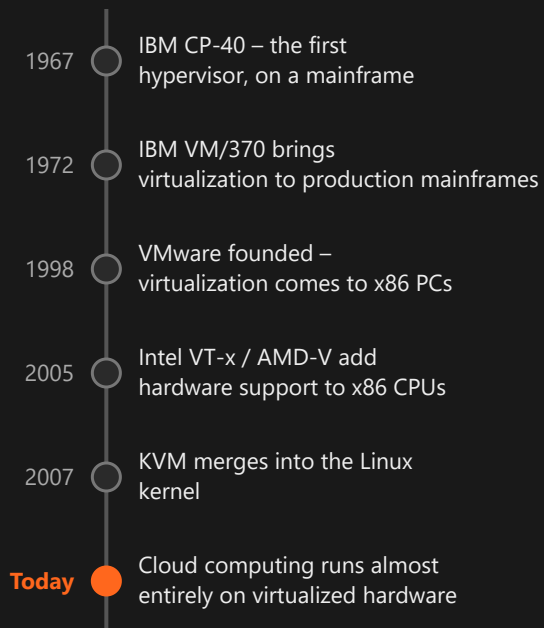
WHERE DOES "VIRTUAL" COME FROM?

- Latin: virtus, power
- In effect, not yet real
- Now: not physical, but real





VIRTUALIZATION





WHERE DOES "HYPERVISOR" COME FROM?

- Hyper-: Greek, over/above
- -visor: short for supervisor
- Together: oversees the OS





WHY USE VIRTUALIZATION?

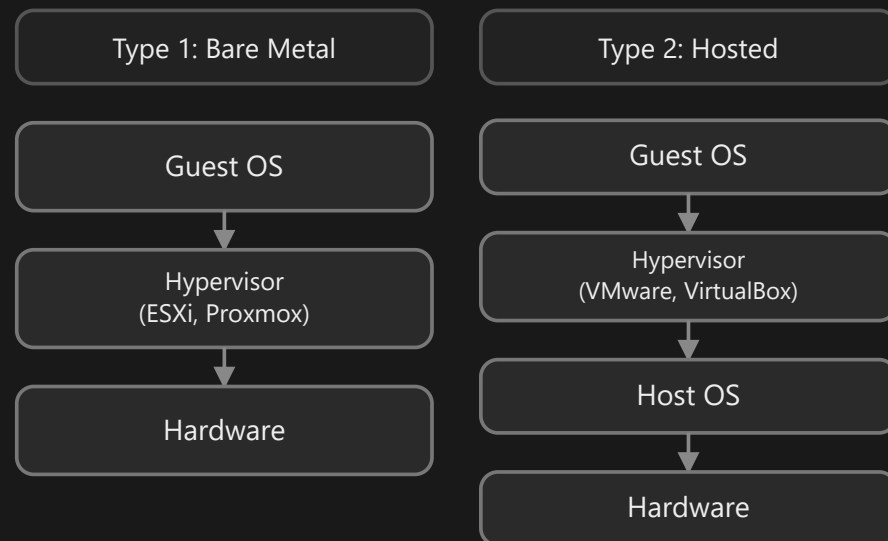
- Isolation, cost savings
- Snapshots & portability
- Better hardware utilization





TYPE 1 VS. TYPE 2 HYPERVISORS

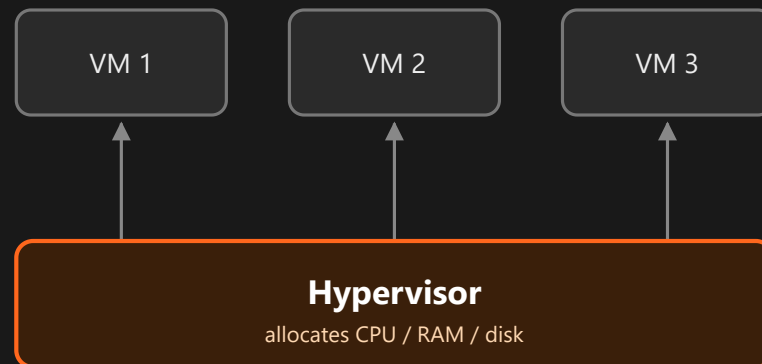
- Type 1: bare-metal (ESXi)
- Type 2: hosted on your OS
- We use Type 2 this course





VIRTUAL MACHINE

- A full computer, virtualized
- Its own OS, CPU, RAM, disk
- Many VMs, one physical box





QUESTIONS?



VIRTUALBOX & WSL



WHERE DO "HOST" & "GUEST" COME FROM?

- Latin: hospes, one word
- Meant host, guest, stranger
- 1972: IBM VM/370 usage





TOP 5 HYPERVISORS

1. VirtualBox (this course)
2. VMware ESXi (vSphere)
3. Microsoft Hyper-V
4. Proxmox VE (KVM-based)
5. XCP-ng (Xen-based)

Hypervisor	Type	License	First Released
VirtualBox	Type 2	Free/open source	2007
VMware ESXi	Type 1	Paid (since 2023)	2001 (as ESX)
Hyper-V	Type 1	Included w/ Windows	2008
Proxmox VE	Type 1	Free/open source (GitHub)	2008
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Popular hypervisor comparison, ranked (Source: [virtualizationhowto.com](https://www.virtualizationhowto.com))



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XCP-ng	Type 1	Free/open source (GitHub)	2018

Popular hypervisor comparison, ranked (Source: [virtualizationhowto.com](https://www.virtualizationhowto.com))

Don't mistake this ranking for market share – VirtualBox is #1 here only because it's what we use.



OUR TOOLSET: VMWARE & VIRTUALBOX

- Both are free to use
- Same concepts, either tool
- Pacific gives VMware access



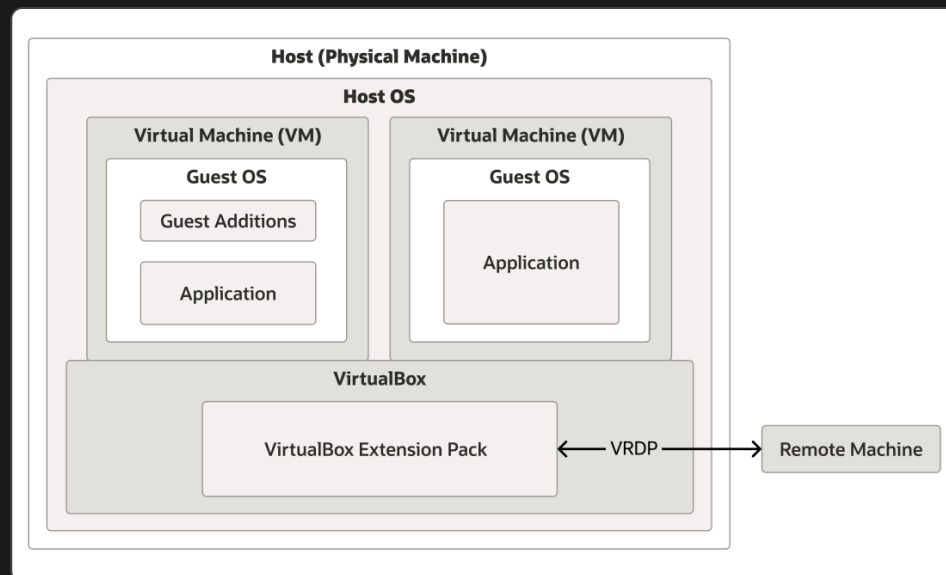
VMware & VirtualBox logos (Source: [Wikipedia](#), [VMware](#); [Wikipedia](#), [VirtualBox](#))

Feature	VMware	VirtualBox
Cost	Free (Player)	Free/open source
Snapshots	Yes	Yes
Networking modes	NAT/Bridged/Host-only	NAT/Bridged/Host-only/Internal



VIRTUALBOX ARCHITECTURE

- One VirtualBox, many VMs
- Each VM: its own Guest OS
- Guest Additions improve it

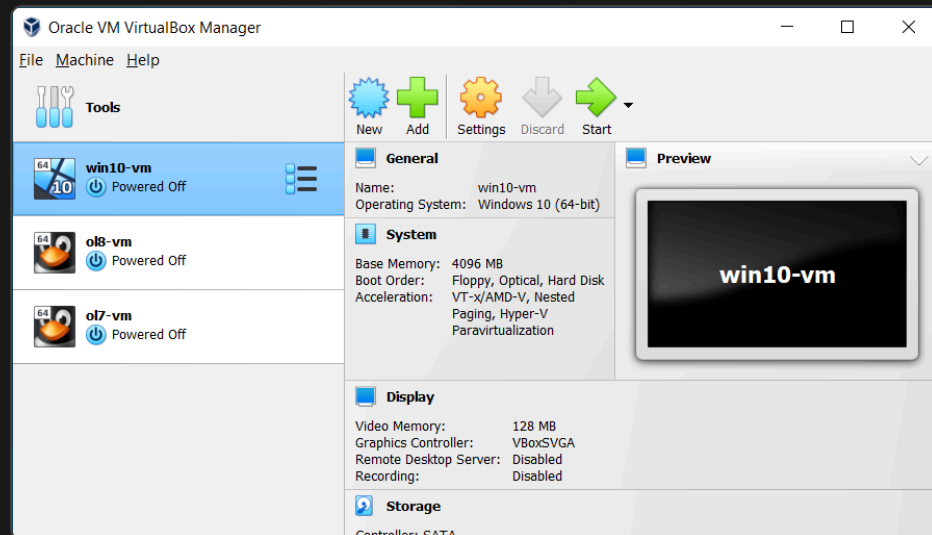


VirtualBox component architecture (Source: [virtualbox.org/manual](https://www.virtualbox.org/manual))



THE VIRTUALBOX MANAGER

- One list of all your VMs
- Start, stop, configure here
- Same app on any host OS

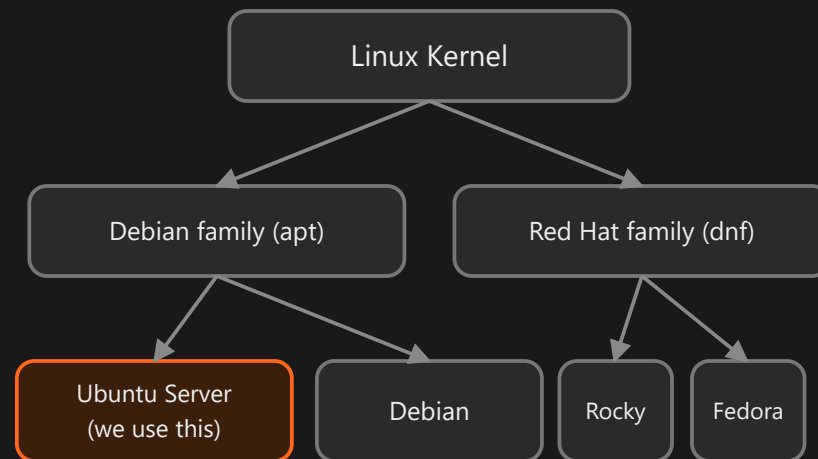


The VirtualBox Manager window (Source: [virtualbox.org/manual](https://www.virtualbox.org/manual))



CHOOSING AN OS: LINUX DISTRIBUTIONS

- We use Ubuntu Server (LTS)
- Debian (apt) vs Red Hat (dnf)
- We standardize on one distro





VIRTUALIZATION IN PRACTICE: WSL2

- Host: Windows 11
- Hypervisor: Hyper-V, built-in
- Guest: Kali Linux kernel
- No VM manager UI needed

Pattern	Example	What You See
Type 1 (bare-metal)	VMware ESXi	Runs directly on hardware
Type 2 (hosted)	VirtualBox	An app on your OS
WSL2 (Microsoft)	Kali on Windows	Auto-managed, no VM UI



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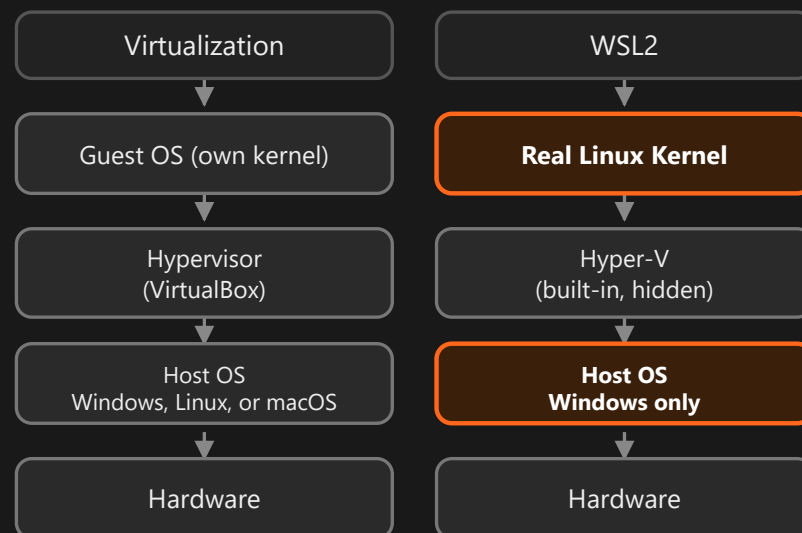
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⚠ WSL2 is still **real** virtualization. A real Linux kernel runs inside a Microsoft-managed VM, using Hyper-V under the hood, even though you never see a VM manager window.



WSL2 VS. VIRTUALIZATION

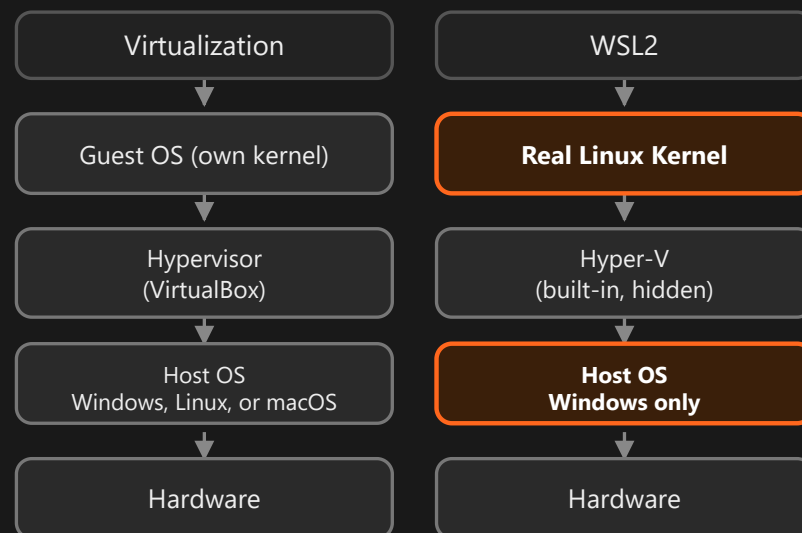
- A real kernel, virtualized
- No visible VM manager
- Runs via Hyper-V, hidden





WSL2 VS. VIRTUALIZATION

- A real kernel, virtualized
- No visible VM manager
- Runs via Hyper-V, hidden



⚠ WSL2 skips a full guest OS. It runs a real Linux kernel **directly**, with no separate virtual BIOS or drivers to install.

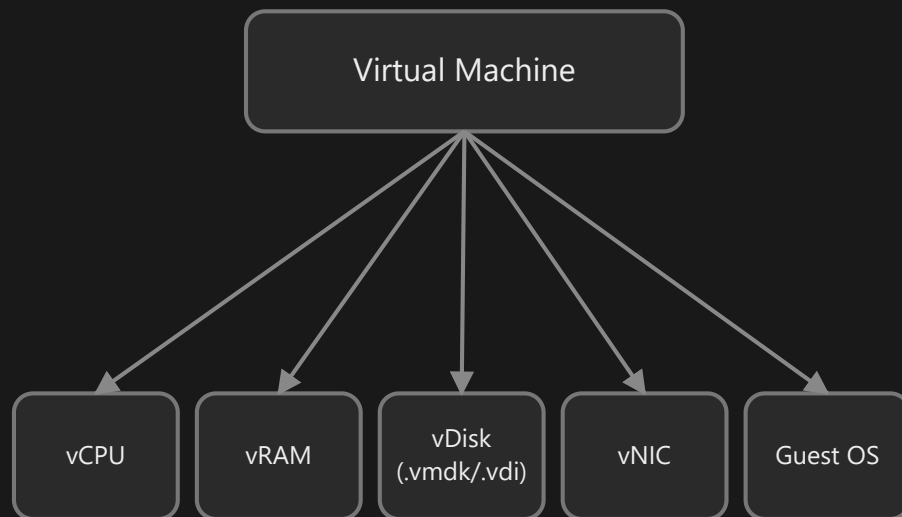


QUESTIONS?



ANATOMY OF A VM

- vCPU: slices of your real CPU
- vRAM: reserved host memory
- vDisk: a file as a hard drive
- vNIC: connects VM to network



Source: [microsoft/WSL, Issue #12567](#)



WHERE DOES "VIRTUAL MACHINE" COME FROM?

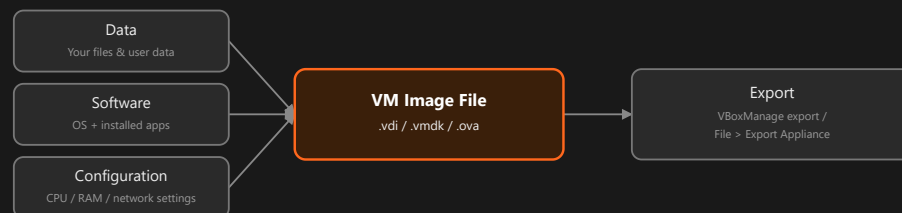
- 1960s: full simulated PC
- 1972: VM/370 ships
- "Virtual machine" sticks





WHAT'S INSIDE A VM IMAGE?

- Data, software, config bundled
- Packed into one image file
- Export as .vdi/.vmdk/.ova





CLI: CREATE A VM

- VBoxManage: VirtualBox's CLI
- Same steps as the GUI wizard
- Scriptable & repeatable

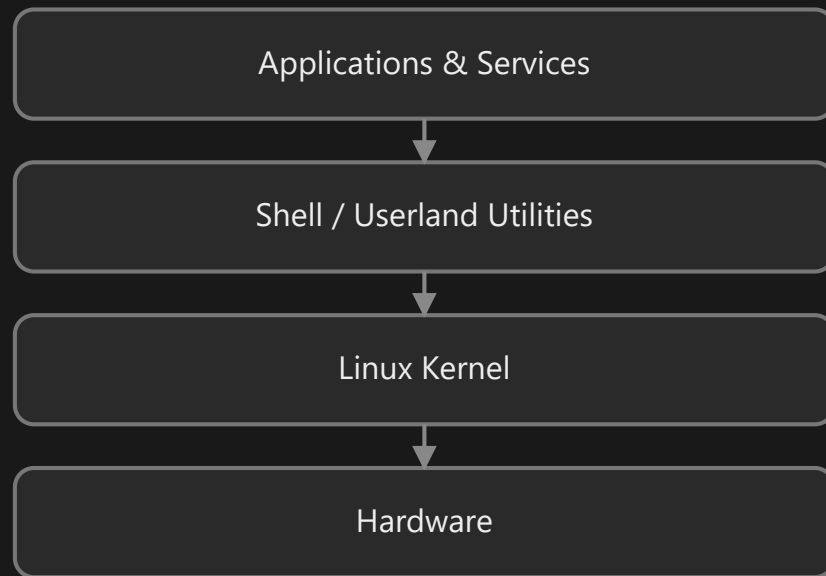
```
$ VBoxManage createvm --name "lab1-ubuntu" \  
    --ostype Ubuntu_64 --register  
  
$ VBoxManage modifyvm "lab1-ubuntu" \  
    --cpus 2 --memory 2048 --nic1 nat  
  
$ VBoxManage createmedium disk \  
    --filename lab1-ubuntu.vdi --size 20000  
$ VBoxManage storagectl "lab1-ubuntu" \  
    --name SATA --add sata
```

[Copy](#)



THE LINUX KERNEL & USERLAND

- Kernel talks to hardware
- Bash talks to the kernel
- Utilities & apps sit on top





WHERE DOES "KERNEL" COME FROM?

- Old English: cyrnel, corn
- Soft part inside a seed
- 1970s: Unix's protected core





VM LIFECYCLE

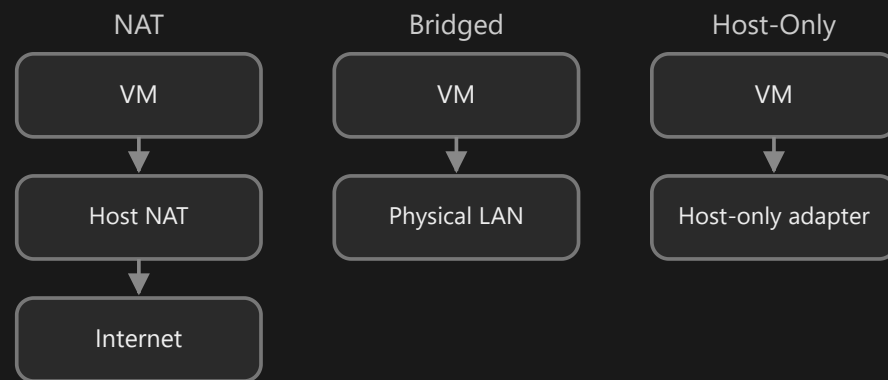
- Create, configure, install OS
- Snapshot before risky changes
- Just a file, destroy freely





VM NETWORKING MODES

- NAT: shares host's IP
- Bridged: own address on LAN
- Host-only: talks to host only
- Mode choice matters per lab





CLI: CONFIGURE NETWORKING

- Check or change NIC mode
- No GUI required
- Sets up automation later

```
$ VBoxManage showvminfo "lab1-ubuntu" \  
    | grep NIC  
  
$ VBoxManage modifyvm "lab1-ubuntu" \  
    --nic1 bridged \  
    --bridgeadapter1 "Wi-Fi"  
  
$ VBoxManage modifyvm "lab1-ubuntu" \  
    --nic1 hostonly \  
    --hostonlyadapter1 vboxnet0
```

Copy



SIZING YOUR VM: CPU, RAM, DISK

- Don't over-allocate resources
- Thin-provision to save space
- Start modest, resize later

Resource	Lab minimum
vCPU	2 cores
vRAM	2 GB
vDisk	20 GB (thin)



WHERE DOES "SNAPSHOT" COME FROM?

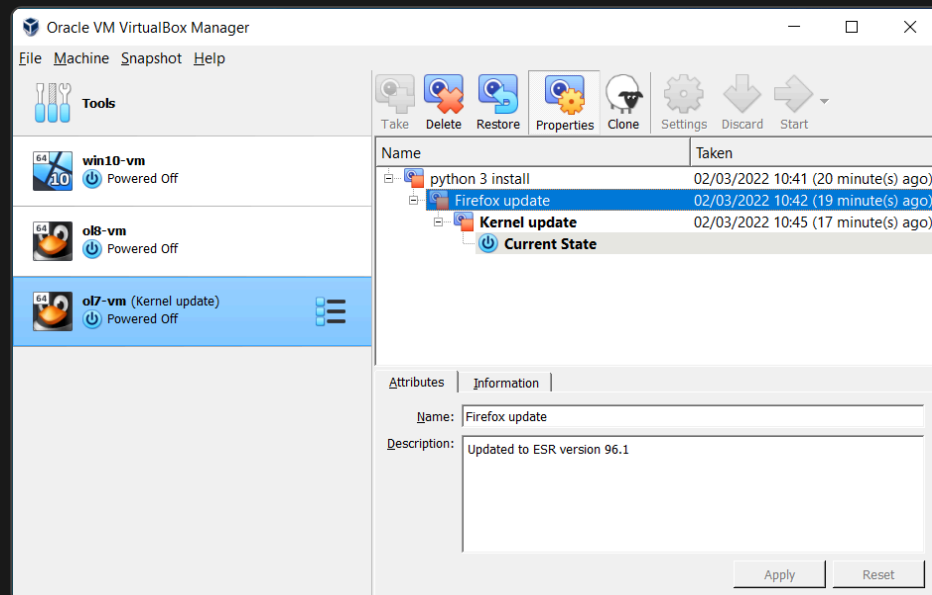
- 1808: hunting, no aim
- 1860: photography borrows it
- Now: VM point-in-time copy





SNAPSHOTS IN THE REAL TOOL

- Right-click a VM → Snapshot
- Named, timestamped, listed
- Restore with one click



Snapshots panel in VirtualBox (Source: [virtualbox.org/manual](https://www.virtualbox.org/manual))



FROM BUILDING A VM TO SECURING IT





QUESTIONS?

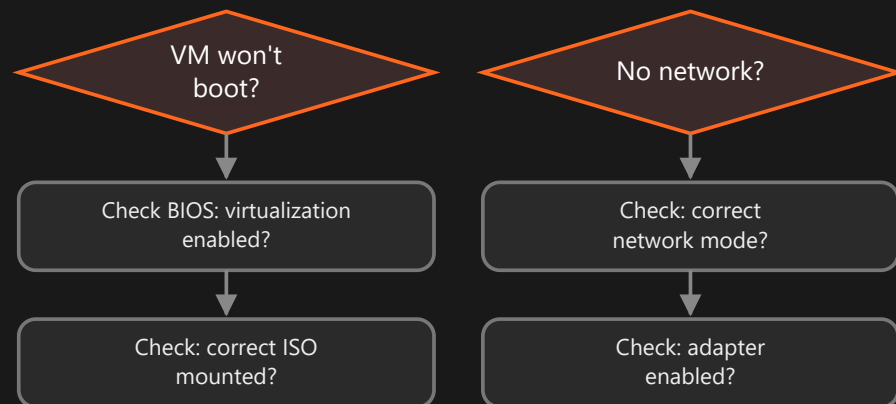


REAL-WORLD RISKS



COMMON SETUP PITFALLS

- VT not enabled in BIOS/UEFI
- Wrong mode blocks networking
- Over-committing RAM/CPU
- Forgetting to snapshot first

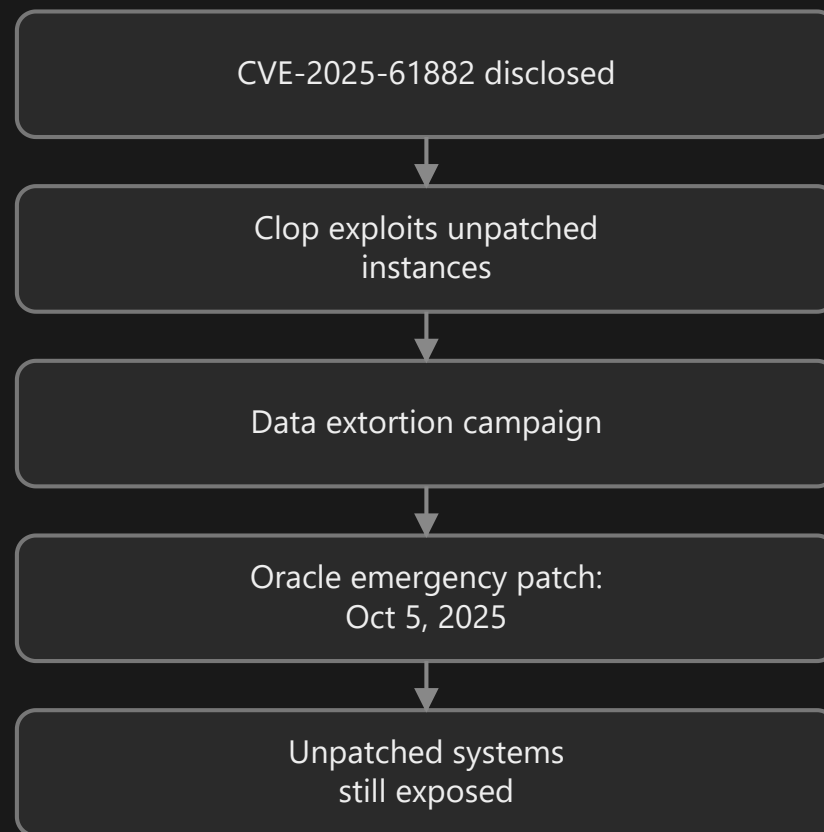




IT'S HAPPENING RIGHT NOW: ORACLE EBS

- CVE-2025-61882: 9.8 zero-day
- Exploited by Clop, Oct 2025
- Patched Oct 5, after exploit
- Unpatched = still exposed

Source: cybersecuritynews.com, infosecurity-magazine.com
(Oct 2025)

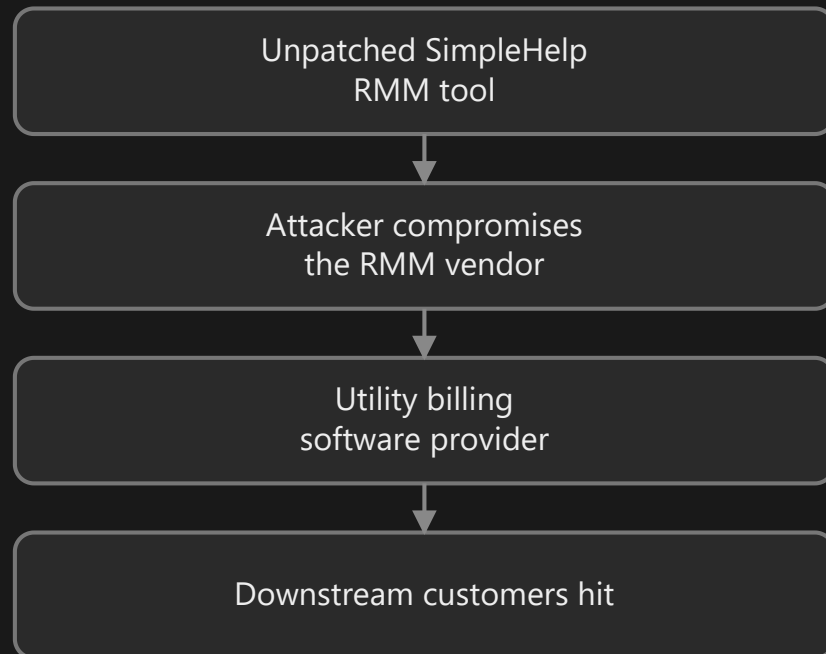




IT'S HAPPENING RIGHT NOW: RMM SUPPLY CHAIN

- Unpatched SimpleHelp RMM tool
- Hit a vendor, its customers
- CISA warned since Jan 2025
- One tool, many victims

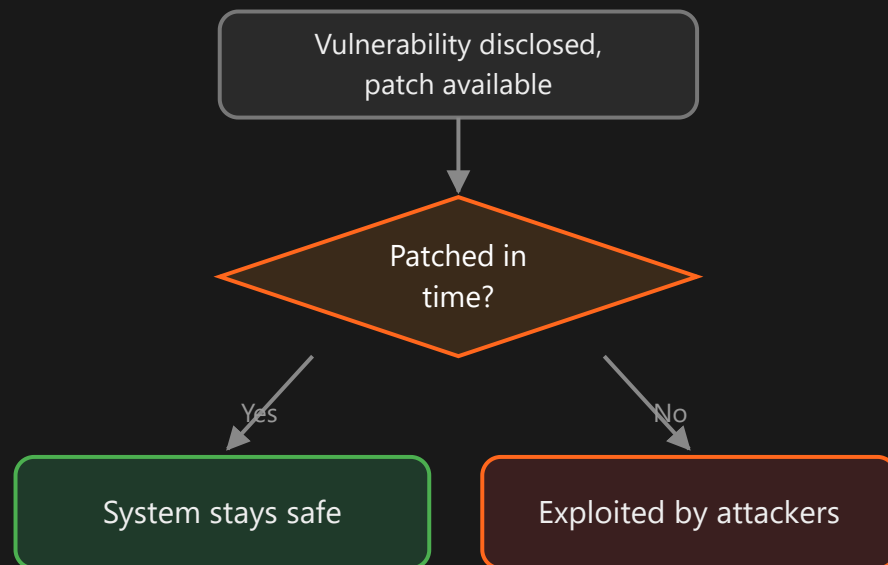
Source: [cisa.gov advisory AA25-163A](https://www.cisa.gov/advisory/AA25-163A) (2025)





SPOT THE PATTERN

- Both used patchable bugs
- Both hit infrastructure
- Spread via trusted tools
- It's a sysadmin problem first





MISCONFIGURATION, BY THE NUMBERS

- 80% of firms breached in 2025
- Misconfig ~ half of breaches
- Exposed ~190 days, unnoticed
- Avg cloud breach: \$4.7M

80% breached in past year

~45-50% caused by misconfig

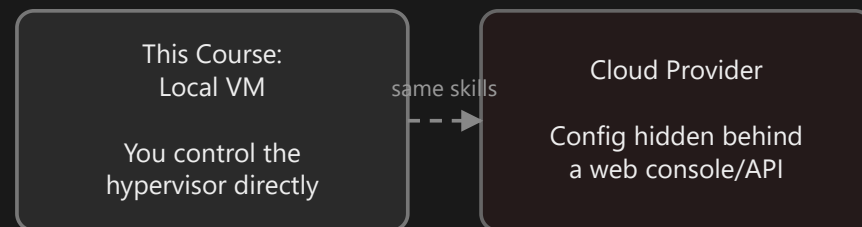
190 days avg. exposure

Source: [fidelissecurity.com](https://www.fidelissecurity.com), datastackhub.com (2025)



WHY LOCAL VMS FOR THIS COURSE

- It's a skill gap, not cloud
- Local VMs give full control
- No billing or cloud quotas
- Skills transfer to any cloud





REVIEW THE STACK

- Foundation: virtualization
- Core skills: files & shells
- Services: DHCP, DNS, logging
- Applied: deploy & automate

M16: Final Presentations & Course Wrap-up

M15: Infrastructure as Code

M14: Containerization

M13: Deployment & Network Segmentation

M12: Firewalls & Network Security

M11: Automation Scripting

M10: Time Synchronization

M9: Logging & Security Monitoring

M8: Name Resolution & Secure Transport

M7: Network Configuration

M6: Service Management

M5: Process Mgmt & Remote Access

M4: Access Control & Auditing

M3: File Systems & Storage

M2: Command-Line & Permissions

M1: Virtualization & Isolation



QUESTIONS?



LAB 1 & HOMEWORK 1



LOOKING AHEAD: MODULE 2

- Core command-line tools
- Search, I/O, permissions
- Lab 2: command-line drills

M16: Final Presentations & Course Wrap-up

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Package	Distro	Shell	GNU	Bash	sudo	Virtualization
Hypervisor	Host / Guest	Virtual Machine	Kernel	Snapshot		

